SPEECH ENHANCEMENT

Theory and Practice

Second Edition

Philipos C. Loizou

CRC Press
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To my parents, wife Demetria, and son Costakis

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Preface to the Second Edition

The second edition of this text not only revises the first edition but also expands it. In particular, it includes two new chapters. Chapter 11 provides a thorough coverage of objective intelligibility measures and Chapter 13 covers algorithms that *can* improve speech intelligibility. The feedback I received from most readers about the first edition is that much of the focus of the text has been placed on algorithms that can improve speech quality rather than speech intelligibility. The second edition comes in response to those readers. With the proliferation of mobile devices and hearing devices (hearing aids and cochlear implants), there is a growing and pressing need to design algorithms that can improve speech intelligibility without sacrificing quality. The inclusion of a chapter on intelligibility measures, which can predict reliably speech intelligibility, is absolutely necessary in order to understand why the algorithms described in Chapter 13 *do* improve speech intelligibility. Secondly, having a good understanding of some of the commonly used intelligibility metrics can assist us in the design of novel noise reduction algorithms that derive statistical estimators that maximize/minimize such metrics. This stands in contrast with the conventional estimators that aim to minimize the mean-square error (MSE), a metric that is "speech ignorant" and not motivated by any known facts about how human listeners are able to perceive speech in complex listening situations. It is the opinion of this author that our obsession with the MSE metric delayed progress in the field of noise reduction.

The contents of the DVD-ROM have also been updated to include (1) MATLAB® code with the implementation of some of the intelligibility measures described in Chapter 11, (2) MATLAB code and C/C++ code with the implementation of the algorithms described in Chapter 13, and (3) real-world noise recordings. In addition, it includes the implementation of the wideband-version of the PESQ measure for assessing the quality of speech sampled at rates higher than 8 kHz.

I wish to express my sincere thanks to all my graduate students who contributed in some way to the writing of the second edition. In particular, I would like to thank my postdoctorate students, Drs. Kamil Wójcicki, Jianfen Ma, and Fei Chen, who wrote many of the MATLAB algorithms included in the updated DVD-ROM. Many thanks also go to Oldooz Hazrati and Dr. Kostas Kokkinakis. Thanks also go to Jacky Gao, Yi Gao, Li Xiao, and Fang Deng who, in the course of translating the first edition to Chinese, found many errors and typos. I am also thankful to Nora Konopka, editor at Taylor & Francis Group, for providing the support and encouragement for this book.

Finally, I would like to express my deepest gratitude to my wife Demetria for her understanding and undying support throughout this project.

Philipos C. Loizou
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Preface to the First Edition

This text is, in part, an outgrowth of my graduate course on speech signal processing, which I have been teaching at the University of Texas at Dallas since the fall of 1999. It is also, in part, a product of my own research in the area. The fact that no textbook existed at the time on speech enhancement, other than a few edited books suitable for the experts, made it difficult to teach the fundamental principles of speech enhancement in a graduate-level course. It must be equally frustrating for new students or speech scientists interested in getting into the field of speech enhancement without having access to a tutorial review or introductory paper (the last review paper was published in the *Proceedings of IEEE* in 1979 by Lim and Oppenheim). That work provided the initial motivation to write this book. My interest in this area stems from my research to develop noise reduction algorithms that can be used to help hearing-impaired listeners (cochlear implant listeners) better communicate in noisy environments.* Crucial to the development of such noise reduction algorithms is the basic understanding of the limitations and potential of existing enhancement algorithms, which I believe this book provides.

The textbook consists of 11 chapters, which are outlined in detail in Chapter 1 (Introduction). It is divided into three main parts. Part I presents the digital-signal processing and speech-signal fundamentals needed to understand speech enhancement algorithms. Part II presents the various classes of speech enhancement algorithms proposed over the past two decades, and Part III presents the methods and measures used to evaluate the performance of speech enhancement algorithms.

The text body is supplemented with examples and figures designed to help the reader understand the theory. The book is accompanied by a DVD-ROM, which contains a speech corpus appropriate for quality and intelligibility evaluation of processed speech, and MATLAB® code with the implementation of major speech enhancement algorithms. It is my strong belief that having access to MATLAB code and a common speech database against which to evaluate new speech enhancement algorithms is crucial and necessary in order to move the field forward. Appendix C provides a detailed description of the contents of the DVD-ROM.

The book can be used as a textbook for a one-semester graduate-level course on speech enhancement. Necessary prerequisites for such a course would be a course on digital signal processing and fundamental knowledge of probability theory, random variables, and linear algebra. This book can also be used as a supplement to an introductory course on speech processing. In this case, Chapters 4 through 8 could be covered along with a select set of sections from Chapters 9 and 10.

I wish to express my sincere thanks to the many colleagues and graduate students who contributed in some way to the writing of this book. I would like to thank Professors Patrick Wolfe, Kuldip Paliwal, Peter Assmann, John Hansen, and

* This work is supported by the National Institutes on Deafness and Other Communication Disorders, NIH.

Jim Kaiser for reviewing chapters of this book and providing valuable feedback. Thanks also go to my graduate students who wrote many of the MATLAB algorithms included in the DVD-ROM. In particular, I would like to thank Sundarrajan Rangachari, Sunil Kamath, Yang Lu, Ning Li, and Yi Hu. Special thanks go to Yi Hu who contributed immensely to this project in the form of ideas for new speech enhancement algorithms (many of which are described in this book) and in terms of implementing those ideas in MATLAB and allowing me to include his code. I would also like to thank Professor John Hansen for allowing me to include some of his MATLAB codes on objective measures and to David Pearce for allowing me to use the noise recordings from the AURORA database. Thanks go to Chaitanya Mamidipally and Jessica Dagley for all their help with the recordings of the noisy speech corpus. I am also thankful to B.J. Clark, acquisitions editor at Taylor & Francis Group, for providing the support and encouragement for this book.

Finally, I would like to express my deepest gratitude to my wife Demetria for her understanding and undying support throughout this project.

Publisher's Note

We were very saddened by Philip's death—such a highly accomplished, intelligent, and gracious man.

Philip passed before he could review the proofs for this book or provide the final files for the DVD. It is with our greatest thanks to Dr. John H.L. Hansen, department head and professor in the Department of Electrical Engineering at The University of Texas at Dallas for diligently reviewing and correcting the page proofs; and to Dr. Kamil Wójcicki for modifying, enhancing, and testing the software that accompanies this book.

Nora Konopka
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In Memoriam



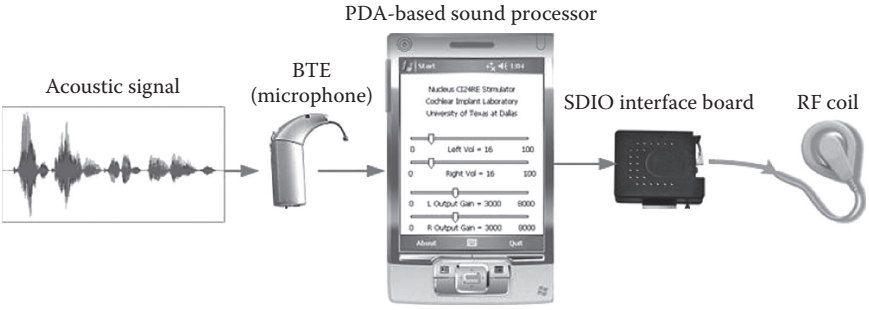
It is with sadness that the Department of Electrical Engineering, University of Texas at Dallas (USA) reports that Professor Philipos Loizou, a professor of Electrical Engineering and pioneer in the fields of hearing aids and speech enhancement, and whose work has helped restore partial hearing to countless people, passed away on Sunday, July 22, 2012 due to cancer. He was 46, and is survived by his wife and 11 year old son.

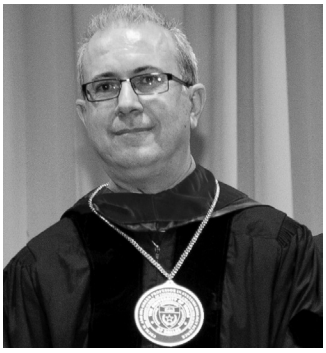
Philipos C. Loizou received the BS, MS, and PhD degrees from Arizona State University (ASU), Tempe, AZ, in 1989, 1991, and 1995, respectively, all in electrical engineering. From 1995 to 1996, he was a postdoctoral fellow in the Department of Speech and Hearing Science, ASU, where he was involved in research related to cochlear implants. From 1996 to 1999, he was an assistant professor at the University of Arkansas at Little Rock, Little Rock, AR. He later moved to join the Department of Electrical Engineering, University of Texas at Dallas, Richardson, TX, where he helped co-found the Center for Robust Speech Systems (CRSS) in the Jonsson School and directed the Speech Processing and Cochlear Implant Labs within CRSS. He served as professor of Electrical Engineering and held the Cecil and Ida Green chair in Systems Biology Science at UT Dallas.



Dr. Loizou was an internationally known leader in signal and speech processing, speech perception and cochlear implant research. He formulated advancements in both signal processing and cochlear implant devices for electrical stimulation of the inner ear of profoundly deaf people. His algorithms also helped improve the performance of cochlear implants by programming the device to operate more effectively in diverse listening environments. More recently, he developed a research platform

that enables smartphones and personal digital assistants (PDAs) to interact with cochlear implants and optimize users' listening experience by controlling stimulation parameters using touch screen user interface. Smartphone application processes the acoustic signal using novel sound processing algorithms and produces electric stimuli in real-time. This has lead to quick development and evaluation of novel research ideas and algorithms for noise suppression, music perception and speech enhancement resulting in better listening experience for implant users. This interface was approved by the U.S. Food and Drug Administration (FDA), and Dr. Loizou was overseeing a clinical trial on the interface with more than a dozen collaborating universities, medical centers, and laboratories.





"He was one of the first persons to explore specific speech enhancement algorithms that directly improve intelligibility—previously believed not to be possible," said Dr. John Hansen, head of the Department of Electrical Engineering, University of Texas at Dallas. "More than his research, Philip was a true scholar—always looking to make contributions which would help improve the quality of life of people with hearing loss."

Loizou attributed much of his success to his students. "I've had very hardworking and dedicated students," he said earlier this year. "Without them, I find it's difficult for me to progress in my research so I owe a lot of praise to them."

He has served as principal investigator (PI/co-PI) on over $9.4M in external funding in his career from U.S. National Institutes of Health, U.S. National Science Foundation, Texas Instruments, Cochlear Corporation, U.T. Southwestern Medical Center, U.S. Air Force, Starkey Corp., Advanced Bionics Corp., and collaborated with many groups, having received the NIH Shannon Award in 1998. He was known for mentoring his students and helping mentor other faculty members in their quest to obtain research support from the NIH.



"He sought out collaborations which would later become some of the most profound contributions in the field," Hansen said. "In addition to his work, he was a valued colleague, mentor and academic citizen. In true Philip style, he always brought a careful, thoughtful approach to all he did and made all around him better."

"There is a lot of research one can do to make cochlear implants more natural for patients, and easier for them to use", was another comment Philip shared with his collaborator Nasser Kehtarnavaz (Professor, Electrical Engineering).

During his career, he graduated 8 PhD and 15 MSEE students. He has published 93 journal papers (primarily in *The Journal of the Acoustical Society of America* and IEEE journals as well as in other journals such as: *Ear and Hearing; American Journal of Audiology; Speech Communication; and Hearing Research*), 54 peer-reviewed conference papers, 5 book chapters, and 3 textbooks including: *Speech Enhancement: Theory and Practice*, Taylor & Francis, Boca Raton, FL (2007); *An Interactive Approach to Signals and Systems Laboratory* (Austin, TX: National Instruments), with co-authors Kehtarnavaz, N. and Rahman, M. (2008); and *Advances in Modern Blind Signal Separation Algorithms: Theory and Applications*, Morgan & Claypool Publishers (with co-author Kokkinakis, K. (2010)).



Dr. Loizou was an elected member of the IEEE Speech-Language Technical Committee (SLTC) (2006–09) for the IEEE Signal Processing Society, and served as member of the Organizing Committee for IEEE ICASSP-2010 (Dallas, TX; USA) overseeing Tutorials. He also served as associate editor during his career for *IEEE Signal Processing Letters* (2006–08), *IEEE Trans. on Speech and Audio Processing* (1999–02), and most recently *IEEE Trans. Biomedical Engineering* and *International Journal of Audiology.* He was elected to the grade of Fellow of the Acoustical Society of America for his work on cochlear implants and speech enhancement.

Dr. Loizou grew up on the Island of Cyprus in the Mediterranean Sea. He enjoyed outdoor activities with his family including hiking, nature, playing/coaching soccer, and especially fishing with his 11-year-old son, Costakis. He is also survived by his wife, Demetria whose background is also in speech and hearing. They were married for 18 years and treasured their common Greek Cypriot culture including dance and regular trips to visit family and friends in Cyprus.

1 Introduction

Speech enhancement is concerned with improving some perceptual aspect of speech that has been degraded by additive noise. In most applications, the aim of speech enhancement is to improve the quality and intelligibility of degraded speech. The improvement in quality is highly desirable as it can reduce listener fatigue particularly in situations where the listener is exposed to high levels of noise for long periods of time (e.g., manufacturing). Speech enhancement algorithms reduce or suppress the background noise to some degree and are sometimes referred to as *noise suppression* algorithms.

The need to enhance speech signals arises in many situations in which the speech signal originates from a noisy location or is affected by noise over a communication channel. There are a wide variety of scenarios in which it is desired to enhance speech. Voice communication, for instance, over cellular telephone systems typically suffers from background noise present in the car, restaurant, etc. at the transmitting end. Speech enhancement algorithms can therefore be used to improve the quality of speech at the receiving end. That is, they can be used as a preprocessor in speech coding systems employed in cellular phone standards (e.g., [1]). If the cellular phone is equipped with a speech recognition system for voice dialing, then recognition accuracy will likely suffer in the presence of noise. In this case, the noisy speech signal can be preprocessed by a speech enhancement algorithm before being fed to the speech recognizer. In an air–ground communication scenario, speech enhancement techniques are needed to improve quality, and preferably intelligibility, of the pilot's speech that has been corrupted by extremely high levels of cockpit noise. In this, as well as in similar communication systems used by the military, it is more desirable to enhance the intelligibility rather than the quality of speech. In a teleconferencing system, noise sources present in one location will be broadcast to all other locations. The situation is further worsened if the room is reverberant. Enhancing the noisy signal prior to broadcasting it will no doubt improve the performance of the teleconferencing system. Finally, hearing impaired listeners wearing hearing aids (or cochlear implant devices) experience extreme difficulty communicating in noisy conditions, and speech enhancement algorithms can be used to somehow preprocess or "clean" the noisy signal before amplification.

The foregoing examples illustrate that the goal of speech enhancement varies depending on the application at hand. Ideally, we would like speech enhancement algorithms to improve both quality and intelligibility. It is possible to reduce the background noise, but at the expense of introducing speech distortion, which in turn may impair speech intelligibility. Hence, the main challenge in designing effective speech enhancement algorithms is to suppress noise without introducing any perceptible distortion in the signal. Thus far, most speech enhancement

algorithms have been found to improve only the quality of speech. The last chapter included in the second edition provides future steps that can be taken to develop algorithms that can improve speech intelligibility.

The solution to the general problem of speech enhancement depends largely on the application at hand, the characteristics of the noise source or interference, the relationship (if any) of the noise to the clean signal, and the number of microphones or sensors available. The interference could be noise-like (e.g., fan noise) or speech-like such as an environment (e.g., restaurant) with competing speakers. Acoustic noise could be additive to the clean signal or convolutive if it originates from a highly reverberant room. Furthermore, the noise may be statistically correlated or uncorrelated with the clean speech signal. The number of microphones available can influence the performance of speech enhancement algorithms. Typically, the larger the number of microphones, the easier the speech enhancement task becomes. Adaptive cancellation techniques can be used when at least one microphone is placed near the noise source.

This book focuses on enhancement of speech signals degraded by statistically uncorrelated (and independent) additive noise. The enhancement algorithms described in this book are not restricted to any particular type of noise, but can be generally applied to a wide variety of noise sources (more on this later). Furthermore, it is assumed that only the noisy signal, containing both the clean speech and additive noise, is available from a single microphone for speech enhancement. This situation constitutes one of the most challenging problems in speech enhancement since it assumes no access to a reference microphone that picks up the noise signal.

1.1 UNDERSTANDING THE ENEMY: NOISE

Prior to designing algorithms to combat additive noise, it is crucial to understand the behavior of various types of noise, the differences between the noise sources in terms of temporal and spectral characteristics, and the range of noise levels that may be encountered in real life.

1.1.1 Noise Sources

We are surrounded by noise wherever we go. Noise is present, for instance, in the street (e.g., cars passing by, street construction work), in the car (e.g., engine noise, wind), the office (e.g., PC fan noise, air ducts), the restaurant (e.g., people talking in nearby tables), and the department stores (e.g., telephone ringing, sales representatives talking). As these examples illustrate, noise appears in different shapes and forms in daily life.

Noise can be stationary, that is, does not change over time, such as the fan noise coming from PCs. Noise can also be nonstationary, such as the restaurant noise, that is, multiple people speaking in the background mixed in some cases with noise emanating from the kitchen. The spectral (and temporal) characteristics of the restaurant noise are constantly changing as people carry on conversations in neighboring tables and as the waiters keep interact and converse with people. Clearly, the task of suppressing noise that is constantly changing (nonstationary) is more difficult than the task of suppressing stationary noise.

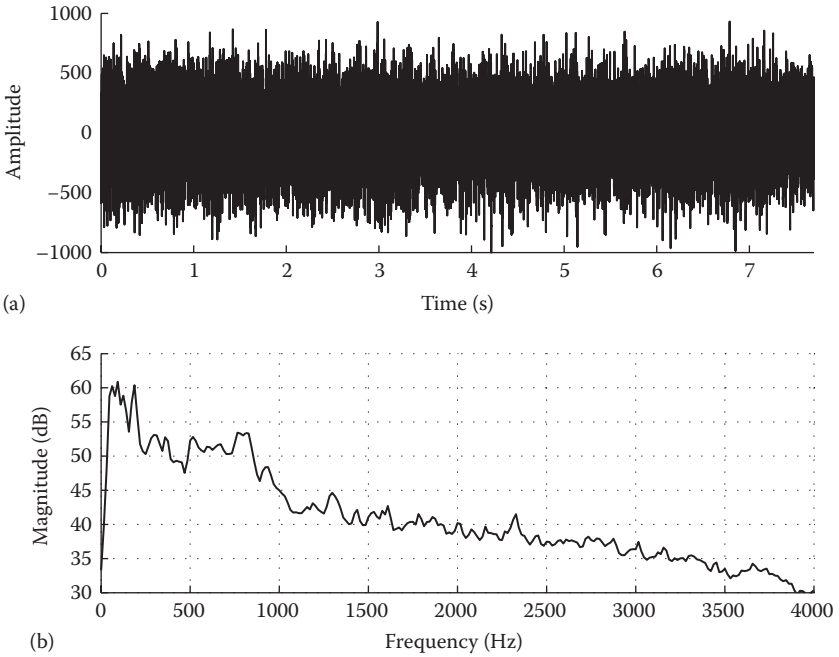


FIGURE 1.1 (a) Example noise from a car and (b) its long-term average spectrum.

Another distinctive feature of the various types of noise is the shape of their spectrum, particularly as it relates to the distribution of noise energy in the frequency domain. For instance, the main energy of wind noise is concentrated in the low frequencies, typically below 500 Hz. Restaurant noise, on the other hand, occupies a wider frequency range. Figures 1.1 through 1.3 show example time waveforms of car noise, train noise, and restaurant noise (noise sources were taken from the NOIZEUS corpus [2], which is included in the DVD of this book). The corresponding long-term average spectra of these noise sources are also shown. In these three example noise sources, the car noise (Figure 1.1) is relatively stationary but the train and restaurant noises are not. It is clear from Figures 1.1 through 1.3 that the differences between these three types of noise sources are more evident in the frequency domain rather than the time domain. Most of energy of the car noise is concentrated in the low frequencies, that is, it is low-pass in nature. The train noise, on the other hand, is more broadband as it occupies a wider frequency range.

1.1.2 Noise and Speech Levels in Various Environments

Critical to the design of speech enhancement algorithms is knowledge of the range of speech and noise intensity levels in real-world scenarios. From that, we can estimate the range of signal-to-noise ratio (SNR) levels encountered in realistic environments. This is important since speech enhancement algorithms need to be effective in suppressing the noise and improving speech quality within that range of SNR levels.

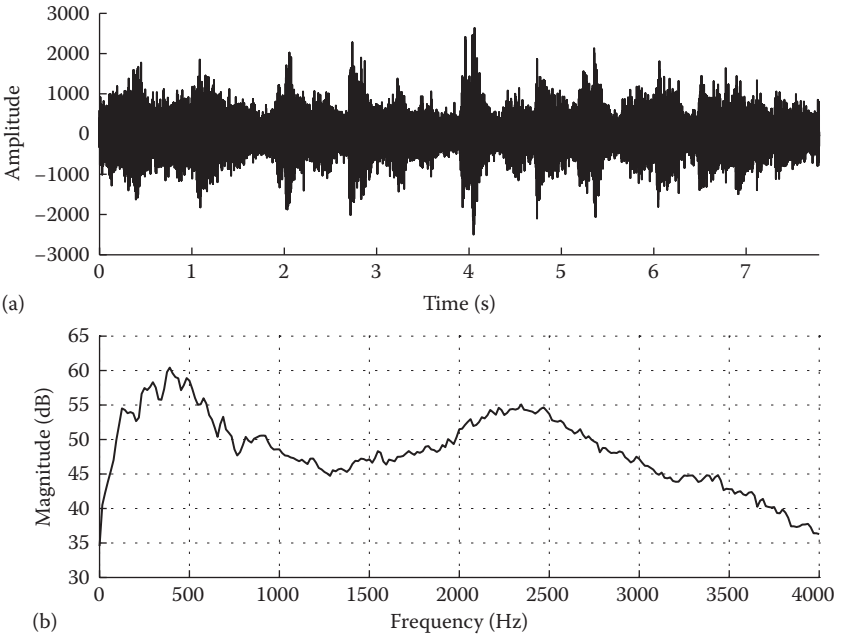


FIGURE 1.2 (a) Example noise from a train and (b) its long-term average spectrum.

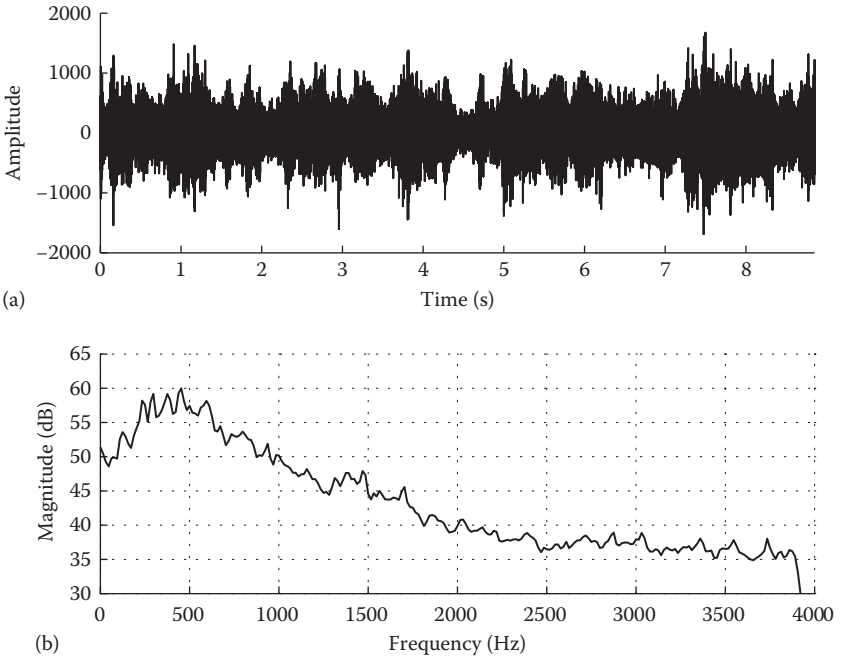


FIGURE 1.3 (a) Example noise from a restaurant and (b) its long-term average spectrum.

A comprehensive analysis and measurement of speech and noise levels in real-world environments was done by Pearsons et al. [3]. They considered a variety of environments encountered in daily life, which included classrooms, urban and suburban houses (inside and outside), hospitals (nursing stations and patient rooms), department stores, trains, and airplanes. The speech and noise levels were measured using sound level meters. The measurements were reported in dB sound pressure level (SPL) (dB SPL is the relative pressure of sound in reference to 0.0002 dynes/cm^2, corresponding to the barely audible sound pressure). As the distance between the speaker and the listener can affect the sound intensity levels, measurements were made with the microphone placed at different distances. Typical distance in face-to-face communication is one meter, and for every doubling of that distance, the sound level is reduced by 6 dB [4]. In certain scenarios (e.g., in transportation vehicles such as trains and airplanes), the distance in face-to-face communication may be reduced to 0.4 m [3].

Figure 1.4 summarizes the average speech and noise levels measured in various environments. Noise levels are the lowest in the classroom, hospital, inside the house, and in the department stores. In these environments, noise levels range between 50 and 55 dB SPL. The corresponding speech levels range between 60 and 70 dB SPL. This suggests that the effective SNR levels in these environments range between 5 and 15 dB. Noise levels are particularly high in trains and airplanes, averaging about 70–75 dB SPL. The corresponding speech levels are roughly the same, suggesting that the effective SNR levels in these two environments are near 0 dB. In classrooms, the effective SNR for students closest to the teacher are most favorable (Figure 1.4), but for those sitting in the back of the classroom it is quite unfavorable as the SNR may approach 0 dB. Noise levels are also high in restaurants and in most cases exceed 65 dB SPL [4]. Generally, it is rare to find a quiet restaurant, as the architectural design of restaurants focuses more on the aesthetics rather than on the acoustics of the interior walls [4]. The study in [5] measured the noise levels in 27 San Francisco Bay Area restaurants and found a median noise level of 72 dBA SPL. The range in noise levels varied across different types of restaurants from a low of

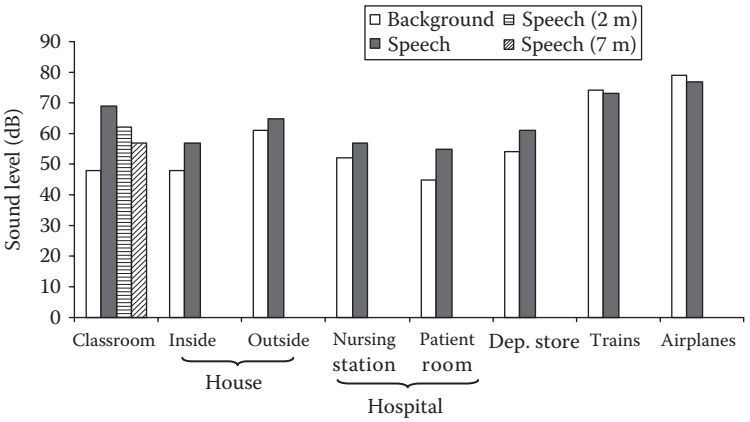


FIGURE 1.4 Average noise and speech levels (measured in dB SPL) in various environments.

59 dBA SPL (family restaurant) to a high of 80 dBA SPL (bistro). In food courts of shopping malls, the range of noise levels varied from 76 to 80 dBA SPL [6].

Note that the speech levels increase in extremely high background noise levels (Figure 1.4). Generally, when the noise level goes beyond 45 dB SPL, people tend to raise their voice levels, a phenomenon known as Lombard effect [7]. The speech level tends to increase with the background level by about 0.5 dB for every 1 dB increase in background level [3]. People stop raising their voice when the ambient noise level goes beyond 70 dB SPL (Figure 1.4). The raising of the voice level is a simple, and efficient, technique that most people use to improve their "listening SNR" whenever in extremely noisy environments.

In brief, for the speech enhancement algorithm to be employed in a practical application it needs to operate, depending on the vocal effort (e.g., normal, raised), within a range of signal-to-noise levels (–5 to 15 dB).

1.2 CLASSES OF SPEECH ENHANCEMENT ALGORITHMS

A number of algorithms have been proposed in the literature for speech enhancement (for an old review of enhancement algorithms, see [8]) with the primary goal of improving speech quality. These algorithms can be divided into three main classes:

1. *Spectral subtractive algorithms*: These are, by far, the simplest enhancement algorithms to implement. They are based on the basic principle that as the noise is additive, one can estimate/update the noise spectrum when speech is not present and subtract it from the noisy signal. Spectral subtractive algorithms were initially proposed by Weiss et al. [9] in the correlation domain *and l*ater by Boll [10] in the Fourier transform domain.
2. *Statistical-model-based algorithms*: The speech enhancement problem is posed in a statistical estimation framework. Given a set of measurements, corresponding say to the Fourier transform coefficients of the noisy signal, we wish to find a linear (or nonlinear) estimator of the parameter of interest, namely, the transform coefficients of the clean signal. The Wiener algorithm and minimum mean square error (MMSE) algorithms, among others, fall in this category. Work in this area was initiated by McAulay and Malpass [11] who proposed a maximum-likelihood approach for estimating the Fourier transform coefficients (spectrum) of the clean signal, followed by the work by Ephraim and Malah [12] who proposed an MMSE estimator of the magnitude spectrum. Much of the work with the Wiener algorithm was initiated in the speech enhancement field by Lim and Oppenheim [8,13].
3. *Subspace algorithms*: Unlike the aforementioned algorithms, the subspace algorithms are rooted primarily on linear algebra theory. More specifically, these algorithms are based on the principle that the clean signal might be confined to a subspace of the noisy Euclidean space. Consequently, given a method of decomposing the vector space of the noisy signal into a subspace that is occupied primarily by the clean signal and a subspace occupied primarily by the noise signal, one could estimate the clean signal simply by

nulling the component of the noisy vector residing in the "noise subspace." The decomposition of the vector space of the noisy signal into "signal" and "noise" subspaces can be done using well-known orthogonal matrix factorization techniques from linear algebra and, in particular, the singular value decomposition (SVD) or the eigenvector–eigenvalue factorization. Work in this area was initiated by Dendrinos et al. [14] who proposed *the u*se of SVD on a data matrix containing time-domain amplitude values and later on by Ephraim and Van Trees [15] who proposed the use of eigenvalue decomposition of the signal covariance matrix.

The aforementioned algorithms have been shown to improve speech quality and reduce listener fatigue, but failed in as far improving speech intelligibility. Recently, a new class of algorithms based on the time-frequency binary mask has been explored and evaluated for noise reduction applications (the binary mask algorithms were originally developed for computational auditory scene analysis and automatic speech recognition applications). The binary mask algorithms are the only algorithms that have been proven to improve speech intelligibility, and constitute the fourth class of algorithms described in this book.

4. *Binary mask algorithms*: Unlike the algorithms in Classes 1–3, which for the most part make use of smooth gain functions for noise suppression, the binary mask algorithms make use of binary gain functions. This amounts to selecting a subset of frequency bins (or channels) from the corrupted speech spectra, while discarding the rest. The selection of those bins is done according to a prescribed rule or criterion. The selection criterion is not unique, and when applied to corrupted speech spectra, can produce large gains in speech intelligibility. Work in this area originated in the field of computational auditory scene analysis [16,17], and for the most part, binary mask algorithms have been utilized in ASR applications [18] rather than for noise reduction. Under ideal conditions (where *a priori* access to the clean *signal* and noise are assumed), the binary mask algorithms can improve speech intelligibility at any SNR level and for any type of background noise. The first demonstration that binary mask algorithms can indeed improve speech intelligibility in binaural conditions was presented in [19] and the first demonstration that binary mask algorithms can improve speech intelligibility in monaural conditions (i.e., where only access to single-microphone noisy observations is available) was presented in [20].

A different chapter is dedicated in this book for each of the aforementioned four classes of algorithms. MATLAB® implementations of representative algorithms from each class are provided in the accompanying DVD.

1.3 BOOK ORGANIZATION

The main goal of this book is to introduce the reader to the basic problems of speech enhancement and the various algorithms proposed to solve these problems. The book is organized into four main parts: "Fundamentals," "Algorithms," "Evaluation," and "Future Steps."

Part I (Fundamentals) consists of Chapters 2 through 4. These chapters provide the fundamentals needed to understand speech enhancement algorithms.

Chapter 2 provides a review of the fundamentals of discrete-time signal processing including discrete-time signals, linear-time invariant systems, z-transform, and Fourier transform representations. Emphasis is placed on the short-time Fourier representation of speech signals, needed to understand many of the speech enhancement algorithms described in later chapters.

Chapter 3 provides a brief overview of the speech production process, properties and acoustic characteristics of speech signals, and the various acoustic cues used by listeners for speech perception. Fundamental knowledge about speech perception is important for designing algorithms capable of improving intelligibility (and quality) in noisy environments.

Chapter 4 presents an overview of the literature on speech perception in noise. It focuses on cues used by human listeners to communicate in highly adverse conditions particularly in situations where multiple talkers are speaking in the background. Understanding the strategies and acoustic cues used by human listeners to segregate speech from other signals in adverse conditions is important as it would help us design better speech enhancement algorithms that would mimic listening strategies used by humans. Such algorithms would hold promise in improving speech intelligibility in noise.

Part II (Algorithms) includes Chapters 5 through 9 and covers all major enhancement algorithms as well as noise estimation algorithms.

Chapter 5 describes the spectral subtractive algorithms for speech enhancement. These algorithms are simple to implement, but suffer for the most part by a so-called musical noise distortion. This chapter describes the many methods proposed to reduce musical noise.

Chapter 6 describes the Wiener filtering approach to speech enhancement. It describes the many variations of Wiener filtering that includes iterative and noniterative methods, and the methods derived from constrained and unconstrained optimization of various error criteria.

Chapter 7 describes statistical-model-based algorithms for speech enhancement. The problem of estimating the spectrum of the clean signal in noisy environments is posed using a statistical estimation framework. This chapter presents various probabilistic-based estimators of the signal spectrum including the maximum likelihood estimator, MMSE, and maximum *a posteriori* estimators.

Chapter 8 describes subspace algorithms for speech enhancement. These algorithms are rooted in linear algebra theory and are based on the principle that the vector space of the noisy signal can be decomposed into "signal" and "noise" subspaces. The decomposition can be done using either the SVD of a Toeplitz-structured data matrix or the eigenvalue decomposition (EVD) of a covariance matrix. This chapter presents SVD-based and EVD-based algorithms for speech enhancement.

Chapter 9 presents noise-estimation algorithms. All speech enhancement algorithms require an estimate of the noise spectrum, and noise estimation algorithms can be used to provide such an estimate. Unlike voice activity detection (VAD) algorithms, noise estimation algorithms estimate and update the noise spectrum

continuously, even during speech activity. As such, noise-estimation algorithms are more suited for speech-enhancement applications operating in highly nonstationary environments.

Part III (Evaluation) consists of Chapters 10 through 12. It provides a description of the measures used to assess the performance (quality and intelligibility) of enhancement algorithms as well as the evaluation results obtained from a comparison between several algorithms.

Chapter 10 provides an overview of the various subjective listening tests used to evaluate speech enhancement algorithms in terms of quality and intelligibility. Human listeners, with normal hearing, participate in these types of tests.

Chapter 11 provides a thorough description of objective quality and intelligibility measures. It also provides the correlation of these measures with human listener's judgments of quality (quality measures) and listener's intelligibility scores (intelligibility measures).

Chapter 12 provides a comprehensive evaluation and comparison of representative speech enhancement algorithms taken from each chapter of this book. It reports on both subjective quality and intelligibility evaluation of speech processed by enhancement algorithms. The quality evaluation is done using subjective listening tests, in which listeners are asked to rate the quality of speech on a five-point scale. The intelligibility evaluation is done with listening tests, wherein listeners are presented with enhanced speech and asked to identify the words spoken.

Part IV (Future Steps) consists of Chapter 13. Under ideal conditions, the algorithms presented in this chapter can improve substantially speech intelligibility, and this chapter provides guidance for future steps that can be taken to realize the full potential of these algorithms under realistic conditions.

Chapter 13 provides a thorough presentation of binary mask algorithms in the context of noise reduction. The full potential of these algorithms is demonstrated in ideal conditions where *a priori* access to the speech and/or noise is assumed. The first steps in implementing these algorithms in realistic conditions are also presented. Given the large benefits in speech intelligibility obtained with these algorithms in various listening conditions, including reverberant conditions, future work is warranted to pursue realistic implementations of binary mask algorithms.

The book is accompanied by a DVD containing speech and noise databases appropriate for evaluation of quality and intelligibility of speech processed by enhancement algorithms. The DVD also contains MATLAB implementations of representative enhancement algorithms taken from each chapter of the book. Detailed description of the DVD contents is provided in Appendix C.

REFERENCES

1. Ramabadran, T., Ashley, J., and McLaughlin, M. (1997), Background noise suppression for speech enhancement and coding, *Proceedings of IEEE Workshop on Speech Coding for Telecommunications*, Pocono Manor, PA, pp. 43–44.
2. Hu, Y. and Loizou, P. (2006), Subjective comparison of speech enhancement algorithms, *Proceedings of IEEE International Conference on Acoustic, Speech, Signal Processing*, Vol. I, Toulouse, France, pp. 153–156.

3. Pearsons, K., Bennett, R., and Fidell, S. (1977), Speech levels in various noise environments, Technical Report EPA-600/1-77-025, U.S. Environmental Protection Agency, Washington, DC.
4. Long, M. (2005), Dinner conversation (an oxymoron?), *Acoust. Today*, 1(1), 25–27.
5. Lebo, C., Smith, M., Mosher, E., Jelonek, S., Schwind, D., Decker, K., Krusemark, H., and Kurz, P. (1994), Restaurant noise, hearing loss, and hearing aids, *West J. Med*., 161(1), 45–49.
6. Porto, M., Navarro, N., and Pimentel, R. (2007), Speech interference in food courts of shopping centers, *Appl. Acoust*., 68, 364–375.
7. Lombard, E. (1911), Le signe de l'élévation de la voix, *Ann. Mal. Oreil. Larynx*., 37, 101–119.
8. Lim, J. and Oppenheim, A. V. (1979), Enhancement and bandwidth compression of noisy speech, *Proc. IEEE*, 67(12), 1586–1604.
9. Weiss, M., Aschkenasy, E., and Parsons, T. (1974), Study and development of the INTEL technique for improving speech intelligibility, Technical Report NSC-FR/4023, Nicolet Scientific Corporation, Northvale, NJ.
10. Boll, S. F. (1979), Suppression of acoustic noise in speech using spectral subtraction, *IEEE Trans. Acoust. Speech Signal Process*., ASSP-27(2), 113–120.
11. McAulay, R. J. and Malpass, M. L. (1980), Speech enhancement using a soft-decision noise suppression filter, *IEEE Trans. Acoust. Speech Signal Process*., ASSP-28, 137–145.
12. Ephraim, Y. and Malah, D. (1984), Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator, *IEEE Trans. Acoust. Speech Signal Process*., ASSP-32(6), 1109–1121.
13. Lim, J. and Oppenheim, A. V. (1978), All-pole modeling of degraded speech, *IEEE Trans. Acoust. Speech Signal Process*., ASSP-26(3), 197–210.
14. Dendrinos, M., Bakamides, S., and Carayannis, G. (1991), Speech enhancement from noise: A regenerative approach, *Speech Commun*., 10, 45–57.
15. Ephraim, Y. and Van Trees, H. L. (1993), A signal subspace approach for speech enhancement, *Proceedings of IEEE International Conference on Acoustic, Speech, Signal Processing*, Vol. II, Minneapolis, MN, pp. 355–358.
16. Brown, G. and Cooke, M. (1994), Computational auditory scene analysis, *Comput. Speech Lang*., 8, 297–336.
17. Weintraub, M. (1985), A theory and computational model of auditory monaural sound separation, PhD thesis, Stanford University, Stanford, CA.
18. Cooke, M., Green, P., Josifovski, L., and Vizinho, A. (2001), Robust automatic speech recognition with missing and uncertain acoustic data, *Speech Commun*., 34, 267–285.
19. Roman, N., Wang, D., and Brown, G. (2003), Speech segregation based on sound localization, *J. Acoust. Soc. Am*., 114, 2236–2252.
20. Kim, G., Lu, Y., Hu, Y., and Loizou, P. (2009), An algorithm that improves speech intelligibility in noise for normal-hearing listeners, *J. Acoust. Soc. Am*., 126(3), 1486–1494.

Part I

Fundamentals

2 Discrete-Time Signal Processing and Short-Time Fourier Analysis

We start with a review of the fundamentals of discrete-time signal processing. Emphasis is placed on the short-time Fourier representation of speech signals, which is needed to understand many of the speech enhancement algorithms described in later chapters. A more thorough coverage of discrete-time signal processing principles can be found in [1,2].

2.1 DISCRETE-TIME SIGNALS

Given a continuous-time (analog) signal, $x_a(t)$, the discrete-time signal $x(n)$ can be obtained by sampling the analog signal at time nT_s, that is,

$$\begin{aligned} x(n) &= x_a(t)\big|_{t=nT_s} \quad -\infty < n < \infty \\ &= x_a(nT_s) \end{aligned} \tag{2.1}$$

where

n is an integer

T_s is the sampling period (s)

The reciprocal of T_s is known as the *sampling frequency*, $F_s = 1/T_s$, expressed in hertz. For perfect signal reconstruction [from discrete $x(n)$ to continuous $x_a(t)$], the Nyquist–Shannon sampling theorem [1] requires that the sampling frequency must be at least twice the highest frequency in the signal.

There are four simple discrete-time signals that are often used as building blocks for constructing more complicated signals. These are the discrete impulse, unit step, exponential, and discrete sinusoid sequences.

The *unit sample* or *discrete impulse* is denoted by $\delta(n)$ and is defined by

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & \text{else} \end{cases} \tag{2.2}$$

The discrete impulse is used as a building block for forming arbitrary discrete-time signals. More specifically, any discrete-time signal can be represented as a sum of weighted and shifted impulses as follows:

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k) \tag{2.3}$$

For example, the discrete-time signal $x(n) = \{1, 0, \underset{\uparrow}{2}, -1, 0, 0, 5\}$, where the arrow indicates the time origin ($n = 0$), can be expressed as

$$x(n) = \delta(n+2) + 2 \cdot \delta(n) - \delta(n-1) + 5 \cdot \delta(n-4) \tag{2.4}$$

The discrete *unit step* function, $u(n)$, is defined by

$$u(n) = \begin{cases} 1 & n \circ 0 \\ 0 & \text{Otherwise} \end{cases} \tag{2.5}$$

and is related to $\delta[n]$ by

$$u(n) = \sum_{k=0}^{\infty} \delta(n-k) \tag{2.6}$$

Note that Equation 2.6 is derived from Equation 2.3, because $x(n) = 1$ for $n \geq 0$. Multiplication of any discrete sequence by $u(n)$ creates a new sequence that is nonzero for $n \geq 0$. Such sequences are called *right-sided* sequences, as they are nonzero in the right-hand side of the time origin. Similarly, multiplication of any discrete sequence by $u(-n)$ creates a new sequence that is nonzero for $n \leq 0$. These sequences are called *left-sided* sequences.

The discrete exponential sequence has the form

$$x(n) = Ca^n \tag{2.7}$$

where C and a are arbitrary constants. The right-sided exponential sequence is given by $x(n) = Ca^n u(n)$ and the left-sided exponential by $x(n) = Ca^n u(-n)$.

Finally, the *discrete-time sinusoid* is given by

$$x(n) = A\cos(2\pi fn + \theta) \quad -\infty < n < \infty \tag{2.8}$$

where

A is the amplitude of the sinusoid

θ is the initial phase

f is the *normalized frequency* and takes values in the range of $-0.5 < f < 0.5$

The discrete-time sinusoid is derived by sampling the continuous (analog) sinusoid as follows:

$$\begin{aligned} x(n) &= A\cos(2\pi Ft + \theta)|_{t=nT_s} \\ &= A\cos\left(2\pi\frac{F}{F_s}n + \theta\right) \\ &= A\cos\left(2\pi fn + \theta\right) \end{aligned} \tag{2.9}$$

where
F is the *frequency* in Hz
f is the *normalized* frequency

It is clear from Equation 2.9 that the two frequencies (F and f) are related by

$$f = \frac{F}{F_s} \tag{2.10}$$

where F_s is the sampling frequency in Hz. Figure 2.1 shows example plots of $\delta(n)$, $u(n)$, $2.5(0.8)^n u(n)$, and $2\cos(2\pi 0.1n)u(n)$.

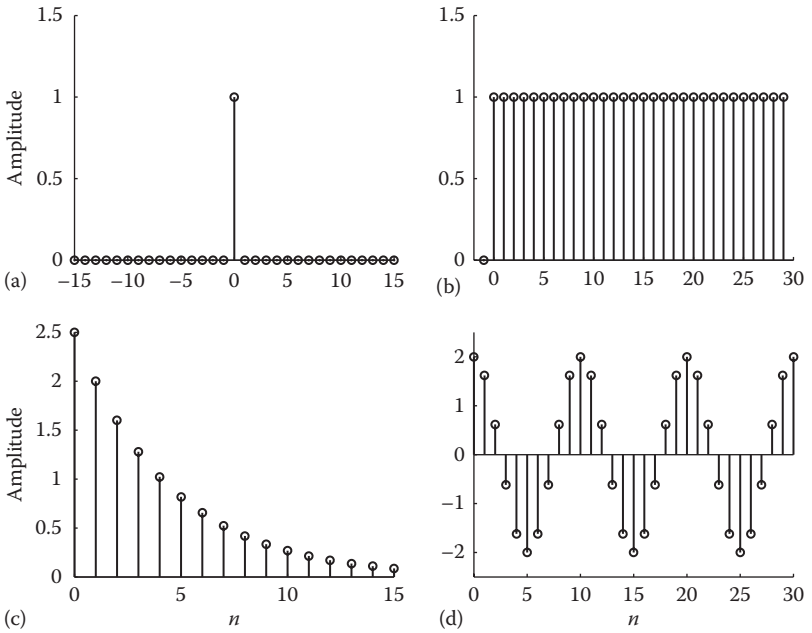


FIGURE 2.1 Example of basic discrete-time signals: (a) $\delta(n)$, (b) $u(n)$, (c) $2.5(0.8)^n u(n)$, and (d) $2\cos(2\pi 0.1n)$.

2.2 LINEAR TIME-INVARIANT DISCRETE-TIME SYSTEMS

Discrete-time systems can be viewed as mathematical transformations that map an input discrete-time signal $x(n)$ into an output discrete-time signal $y(n)$. We denote these transformations by

$$y(n) = T\{x(n)\} \tag{2.11}$$

where the mapping function $T\{\cdot\}$ can be linear or nonlinear. In this chapter, we restrict ourselves to systems that are both linear and time invariant (LTI). Linear systems satisfy the superposition principle, and time-invariant systems possess the following property: $T\{x(n-m)\} = y(n-m)$; that is, a system is said to be time invariant if a shift in the input signal by m samples [i.e., $x(n-m)$] results in a shift in the output by m samples [i.e., $y(n-m)$].

2.2.1 Difference Equations

LTI systems are often described in terms of difference equations. For instance, the following difference equation describes the relationship between the discrete-time input $x(n)$ and the output signal $y(n)$:

$$y(n) = 2x(n) + 0.7y(n-1) \tag{2.12}$$

The output signal at time index n is obtained by multiplying the input signal by 2 and adding the result to the previous output value, which is multiplied by 0.7. If we assume that the input signal is a discrete step function, that is, $x(n) = u(n)$, then we can compute the first three values of the output signal $y(n)$, $n = 0, 1, 2$ as follows:

$$\begin{aligned} y(0) &= 2 && \dagger n = 0 \\ y(1) &= 2 + 0.7 \cdot 2 = 3.4 && n = 1 \\ y(2) &= 2 + 0.7 \cdot (3.4) = 4.38 && n = 2 \end{aligned} \tag{2.13}$$

assuming zero initial conditions, that is, $y(n) = 0$, $n < 0$. Values of $y(n)$ for $n > 2$ can be obtained in a similar fashion following the recursion in Equation 2.12. It is clear that the system in Equation 2.12 is a function of the input signal as well as the past output signals. It is, therefore, recursive in nature. The output signal $y(n)$ in the preceding example is known as the *step response*, as it is the response of the system to a step input function. Similarly, the response of a system to an impulse is known as *impulse response,* often denoted by $h(n)$; that is, $h(n) = T\{\delta(n)\}$.

The impulse response provides a great deal of information about the underlying system. More specifically, the impulse response provides information about the stability and causality of a system. A system is said to be *causal* if

$$h(n) = 0, \quad n < 0 \tag{2.14}$$

In a causal system, the output at time n depends only on the present and the past and does not depend on future samples. Causal systems are desirable as they are amenable to real-time implementation. A discrete-time system is *stable* if

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \tag{2.15}$$

That is, the impulse response has to be absolutely summable.

The general equation describing LTI systems is given by

$$y(n) = \sum_{k=0}^{M-1} b_k x(n-k) + \sum_{k=1}^{N} a_k y(n-k) \tag{2.16}$$

It is clear that the difference equation in Equation 2.12 is a special case of Equation 2.16, with $b_0 = 2$ and $a_1 = 0.7$.

Systems with $a_k = 0$ ($k = 1, 2, \ldots, N$) in Equation 2.16 are known as *finite impulse response* (FIR) systems, and systems with $b_k \neq 0$ and $a_k \neq 0$ (for all k) are known as *infinite impulse response* (IIR) systems. FIR systems have finite-duration impulse responses, whereas IIR systems have infinite-duration impulse responses.

2.2.2 Linear Convolution

The general equation relating the output of an LTI system to a given input sequence $x(n)$ is given by the convolution sum:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \tag{2.17}$$

where $h(n)$ is the impulse response of the system. For simplicity, we often use the following notation for the convolution:

$$y(n) = x(n) * h(n) \tag{2.18}$$

where $*$ denotes the convolution operator. The convolution operation is commutative, associative, and distributive over addition [1]. The convolution operation distributes over addition such that

$$x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n) \tag{2.19}$$

where $h_1(n)$ and $h_2(n)$ are impulse responses. This property suggests that a system connected in parallel with impulse responses $h_1(n)$ and $h_2(n)$ is equivalent to a single system whose impulse response is the sum of the individual impulse responses.

2.3 *z*-TRANSFORM

The z-transform is an extremely valuable tool that can be used to analyze discrete-time signals and systems. The z-transform of a sequence $x(n)$ is defined by

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \tag{2.20}$$

where z is a complex variable. The set of z values for which $X(z)$ converges is called the *region of convergence* (ROC). The ROC and $X(z)$ are both required to uniquely determine the sequence $x(n)$.

2.3.1 PROPERTIES

The z-transform has a number of properties that are often used in the analysis of discrete-time signals and systems. We list in the following text a few of those properties. Table 2.1 shows the other properties of the z-transform. We denote the relationship between the sequence $x(n)$ and its z-transform $X(z)$ by

$$x(n) \overset{Z}{\Leftrightarrow} X(z) \tag{2.21}$$

or equivalently $X(z) = Z[x(n)]$, where $Z[\cdot]$ denotes the z-transformation. The time-shifting and convolution properties are widely used in the analysis of discrete-time signals and are described in the following text.

Let $x(n)$ be a sequence defined only for $n \geq 0$, then

$$Z[x(n - n_0)] = z^{-n_0} X(z) \tag{2.22}$$

where n_0 corresponds to the time delay in samples. This property is called the *time-shifting* property of the z-transform, which is often used to convert difference

TABLE 2.1
Properties of the *z*-Transform

Property	Sequence	*z*-Transform
Linearity	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(z) + a_2X_2(z)$
Time shift	$x(n - n_0)$	$z^{-n_0}X(z)$
Frequency scaling	$a^n x(n)$	$X\frac{z}{a}$
Multiplication by n	$nx(n)$	$-z\frac{d}{dz}X(z)$
Convolution	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$

equations in the z-domain. For instance, the difference equation in Equation 2.12 can be converted in the z-domain using the shifting property (Equation 2.22) and the linearity property as follows:

$$Y(z) = 2X(z) + 0.7z^{-1}Y(z) \tag{2.23}$$

The convolution property relates the convolution of two sequences in terms of their z-transforms. Let $x_1(n)$ and $x_2(n)$ be two sequences with the following z-transforms and ROCs:

$$\begin{aligned} x_1(n) &\overset{Z}{\Leftrightarrow} X_1(z) \quad \text{ROC} = R_1 \\ x_2(n) &\overset{Z}{\Leftrightarrow} X_2(z) \quad \text{ROC} = R_2 \end{aligned} \tag{2.24}$$

where R_1 and R_2 are the ROCs of $x_1(n)$ and $x_2(n)$, respectively. Then, the z-transform of the convolution of the two sequences is given by

$$x_1(n) * x_2(n) \overset{Z}{\Leftrightarrow} X_1(z) \cdot X_2(z) \tag{2.25}$$

and the resulting ROC is given by the intersection of the individual ROCs, that is, $R_1 \cap R_2$. Hence, convolution in the time domain is equivalent to multiplication in the z-domain. For that reason, it is sometimes easier, and preferred, to operate in the z-domain than in the time domain.

2.3.2 z-Domain Transfer Function

Making use of the convolution property of the z-transform, we can analyze LTI systems in the z-domain. More specifically, the relationship between the input sequence $x(n)$ and output sequence $y(n)$ in Equation 2.18 can be expressed as

$$Y(z) = H(z)X(z) \tag{2.26}$$

where $Y(z)$, $H(z)$, and $X(z)$ are the z-transforms of the output sequence, impulse response, and input sequence, respectively. $H(z)$ is known as the discrete-time *system function* or *transfer function* as it relates the output to the input.

We can derive the transfer function of the general constant-coefficient difference equation (Equation 2.16) by using the linearity and shifting properties:

$$Y(z) = X(z)\sum_{k=0}^{M-1} b_k z^{-k} + Y(z)\sum_{k=1}^{N} a_k z^{-k} \tag{2.27}$$

Solving Equation 2.27 for $H(z) = Y(z)/X(z)$, we get

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{1 - \sum_{k=1}^{N} a_k z^{-k}} \tag{2.28}$$

After computing the roots of the polynomials in the numerator and denominator, we can express Equation 2.28 in factored form as

$$H(z) = b_0 \frac{\prod_{k=1}^{M-1} \left(1 - c_k z^{-1}\right)}{\prod_{k=1}^{N-1} \left(1 - p_k z^{-1}\right)} \tag{2.29}$$

where

c_k are the roots [also called the *zeros* of $H(z)$] of the numerator polynomial
p_k are the roots [also called *poles* of $H(z)$] of the denominator polynomial

This form of the transfer function is often used to check for system stability. If the poles are inside the unit circle, that is, $|p_k| < 1$, then the system is bounded-input bounded-output stable, otherwise it is considered unstable.

As an example, we consider the system given in Equation 2.12. From Equation 2.23, we can compute the system transfer function as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2}{1 - 0.7z^{-1}} \tag{2.30}$$

From the preceding equation, we can see that $H(z)$ has a pole at $z = 0.7$ and a zero at $z = 0$. Because the pole is smaller than one, the system is stable. Note that the impulse response of the system, that is, $h(n)$, can be computed using the inverse z-transform of $H(z)$. After making use of the following z-transform pair [2]

$$a^n u(n) \overset{Z}{\leftrightarrow} \frac{1}{1 - az^{-1}} \quad \text{ROC:} |z| > a \tag{2.31}$$

we get

$$h(n) = 2(0.7)^n u(n) \tag{2.32}$$

for the impulse response of the system given in Equation 2.12.

2.4 DISCRETE-TIME FOURIER TRANSFORM

The discrete-time Fourier transform (DTFT) of a sequence $x(n)$ is given by

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \tag{2.33}$$

It is easy to show that $X(\omega) = X(\omega + 2\pi)$, that is, $X(\omega)$ is periodic with a period of 2π. Given $X(\omega)$, we can recover the sequence $x(n)$ using the inverse DTFT:

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} \, d\omega \tag{2.34}$$

In general, the DTFT is a complex-valued function of ω and can be represented in polar form as

$$X(\omega) = |X(\omega)| e^{j\phi_x(\omega)} \tag{2.35}$$

where

$|X(\omega)|$ is the *magnitude spectrum*

$\phi_x(\omega)$ is the *phase spectrum*

For real discrete sequences, $X(\omega)$ is conjugate symmetric. This means that the magnitude spectrum is an even function of ω, that is, $|X(\omega)| = |X(-\omega)|$, and the phase spectrum is an odd function of ω, that is, $\phi_x(\omega) = -\phi_x(-\omega)$.

The DTFT, $X(\omega)$, is generally complex for real-valued signals that have no symmetry. If, however, a signal $x(n)$ is even [i.e., $x(n) = x(-n)$], then $X(\omega)$ is purely real (see Example 2.1). If a signal $x(n)$ has odd symmetry [$x(n) = -x(-n)$], then $X(\omega)$ is purely imaginary.

Example 2.1: Compute the DTFT of the Signal:

$$x(n) = \begin{cases} 1 & -M \le n \le M \\ 0 & \text{Otherwise} \end{cases} \tag{2.36}$$

Using the DTFT definition (Equation 2.33), we get

$$\begin{aligned} X(\omega) &= \sum_{n=-M}^{M} e^{-j\omega n} \\ &= \sum_{n=-M}^{-1} e^{-j\omega n} + \sum_{n=0}^{M} e^{-j\omega n} \end{aligned} \tag{2.37}$$

These summations can be evaluated in closed form using the following identity:

$$\sum_{n=0}^{N} a^n = \frac{1 - a^{N+1}}{1 - a} \tag{2.38}$$

Using Equation 2.38 in Equation 2.37 and after some manipulation, we get

$$X(\omega) = \frac{e^{j\omega} - e^{j\omega(M+1)}}{1 - e^{j\omega}} + \frac{1 - e^{-j\omega(M+1)}}{1 - e^{-j\omega}}$$
$$= \frac{\sin\left[\left(M + \frac{1}{2}\right)\omega\right]}{\sin(\omega/2)} \tag{2.39}$$

Note that $X(\omega)$ is real because $x(n)$ has even symmetry. The magnitude and phase spectra are given by

$$|X(\omega)| = \left|\frac{\sin\left[\left(M + \frac{1}{2}\right)\omega\right]}{\sin(\omega/2)}\right|$$
$$\theta_x(\omega) = \begin{cases} 0 & \text{if } X(\omega) > 0 \\ \pi & \text{if } X(\omega) < 0 \end{cases} \tag{2.40}$$

The magnitude and phase spectra given in the preceding text are plotted in Figure 2.2.

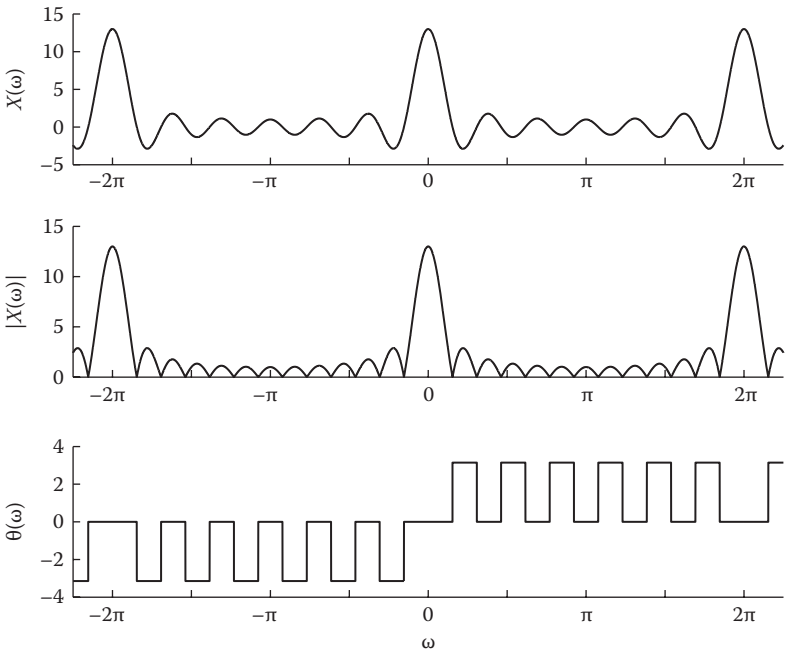


FIGURE 2.2 Magnitude and phase spectra of a symmetric rectangular function (Example 2.1).

2.4.1 DTFT Properties

The DTFT has a number of properties, a few of which we list in the following text. Table 2.2 lists some additional properties of the DTFT. We denote the relationship between the sequence $x(n)$ and its DTFT by

$$x(n) \overset{DTFT}{\Leftrightarrow} X(\omega) \tag{2.41}$$

The *time-shifting* property states that if

$$x(n) \overset{DTFT}{\Leftrightarrow} X(\omega) \tag{2.42}$$

then

$$x(n-n_0) \overset{DTFT}{\circ} e^{-j\omega n_0} X(\omega) \tag{2.43}$$

where n_0 is an integer corresponding to the delay in samples.

The *modulation property* states that multiplying a sequence by a complex exponential corresponds to shifting its DTFT in the frequency domain, that is,

$$e^{j\omega_0 n} x(n) \overset{DTFT}{\circ} X(\omega - \omega_0) \tag{2.44}$$

The *multiplication property* states that multiplication of two sequences results in a periodic convolution of their DTFTs. That is, if

$$\begin{aligned} x(n) &\overset{DTFT}{\Leftrightarrow} X(\theta) \\ w(n) &\overset{DTFT}{\Leftrightarrow} W(\theta) \end{aligned} \tag{2.45}$$

TABLE 2.2
Properties of DTFT

Property	Sequence	DTFT Transform
Linearity	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(\omega) + a_2X_2(\omega)$
Time shift	$x(n-n_0)$	$e^{-j\theta n_0}X(\theta)$
Modulation	$e^{-j\theta_0 n}x(n)$	$X(\omega-\omega_0)$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\theta)X_2(\omega-\theta)d\theta$
Convolution	$x_1(n)*x_2(n)$	$X_1(\omega)X_2(\omega)$

then

$$x(n)w(n) \overset{DTFT}{\Leftrightarrow} \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)W(\omega-\theta)\,d\theta \tag{2.46}$$

The *convolution property* states that the convolution of two sequences results in multiplication of the corresponding DTFTs. For the two sequences given in Equation 2.45, we have

$$x(n) * w(n) \overset{DTFT}{\Leftrightarrow} X(\omega)W(\omega) \tag{2.47}$$

In the context of LTI systems, if $h(n)$ is the impulse response of the system, its DTFT, that is,

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} \tag{2.48}$$

is known as the *frequency response* of the LTI system.

Finally, *Parseval's theorem* relates the total energy in the sequence to its DTFT. It states that the total energy of the sequence $x(n)$, defined as $\Sigma_n|x(n)|^2$, is related to its DTFT according to

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \tag{2.49}$$

2.4.2 Discrete Fourier Transform

The DTFT of a discrete-time sequence is a continuous function of frequency ω (see Equation 2.33). In practice, the sequence $x(n)$ is finite in duration (e.g., consisting of N samples), and we can sample the DTFT at N uniformly spaced frequencies, that is, at $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$. This sampling yields a new transform referred to as the *discrete Fourier transform* (DFT). The DFT of $x(n)$ is given by

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}} \quad 0 \le k \le N-1 \tag{2.50}$$

Given $X(k)$, we can recover $x(n)$ from its DFT, using the inverse DFT (IDFT):

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi kn}{N}} \quad 0 \le n \le N-1 \tag{2.51}$$

Many of the properties of the DFT are similar to those of the DTFT. The main distinction is that many of the operations with the DFT are done modulo N. This is largely due to the implicit periodicity of the sequence $x(n)$ [1, Chapter 5].

We will consider the convolution properties of the DFT (a list of other properties of the DFT can be found in [1, Chapter 5]). We saw earlier (see Equation 2.46) that multiplication of two signals corresponds to a convolution of the individual DTFTs. In the DFT domain, however, multiplication of the two discrete-time sequences does not correspond to *linear convolution* but *circular convolution* of the individual DFTs. That is,

$$x(n)w(n) \overset{DFT}{\Leftrightarrow} X(k) \otimes W(k) = \frac{1}{N} \sum_{m=0}^{N-1} X(m)W((k-m)_N) \tag{2.52}$$

where $\otimes$ denotes circular convolution, $X(k)$ and $W(k)$ are the N-point DFTs of $x(n)$ and $w(n)$, respectively, and the notation $(i)_j$ indicates integer i modulo integer j.

Multiplication of the DFTs of two sequences does not correspond to linear convolution but circular convolution of the two sequences, that is,

$$x(n) \otimes w(n) = \sum_{m=0}^{N-1} x(m)W((n-m)_N) \overset{DFT}{\Leftrightarrow} X(k)W(k) \tag{2.53}$$

Therefore, we cannot use the inverse DFT of $X(k)W(k)$ to obtain the linear convolution of $x(n)$ with $w(n)$. In other words, $IDFT\{X(k)W(k)\} \neq x(n) * w(n)$, where $IDFT\{\cdot\}$ denotes the inverse DFT operation (Equation 2.51).

It is possible, however, to use the DFT to compute the linear convolution of two sequences provided we properly pad the two sequences with zeros. Suppose that $x(n)$ is nonzero over the interval $0 \leq n \leq M-1$ and $w(n)$ is nonzero over the interval $0 \leq n \leq L-1$. If we choose an N-point DFT such that $N \geq M+L-1$, the circular convolution will be equivalent to linear convolution; that is, $IDFT\{X(k)W(k)\} = x(n) * w(n)$. The requirement that $N \geq M+L-1$ comes from the fact that the length of the output sequence resulting from linear convolution is equal to the sum of the individual lengths of the two sequences minus one [1, Chapter 2].

Figure 2.3 shows an example of circular convolution of two discrete-time sequences, a triangle sequence $x(n)$, $n = 0, 1, \ldots, 9$ (Figure 2.3a) and a truncated ramp sequence $h(n)$, $n = 0, 1, \ldots, 15$ (Figure 2.3b). Figure 2.3c shows the output of the circular convolution of the two sequences obtained by computing the IDFT of $X(k)H(k)$, where $X(k)$ denotes the 16-point DFT of the triangle sequence, and $H(k)$ denotes the 16-point DFT of the truncated ramp sequence. Figure 2.3d shows the output of the linear convolution of the two sequences obtained by computing the IDFT of $X_p(k)H_p(k)$, where $X_p(k)$ denotes the 25-point DFT of the triangle sequence, and $H_p(k)$ denotes the 25-point DFT of the truncated ramp sequence. $X_p(k)$ was obtained by padding the triangle sequence with 15 zeros, whereas $H_p(k)$ was obtained by padding the truncated ramp sequence with 9 zeros. The circular convolution effects are evident when comparing the outputs shown in Figure 2.3c and d.

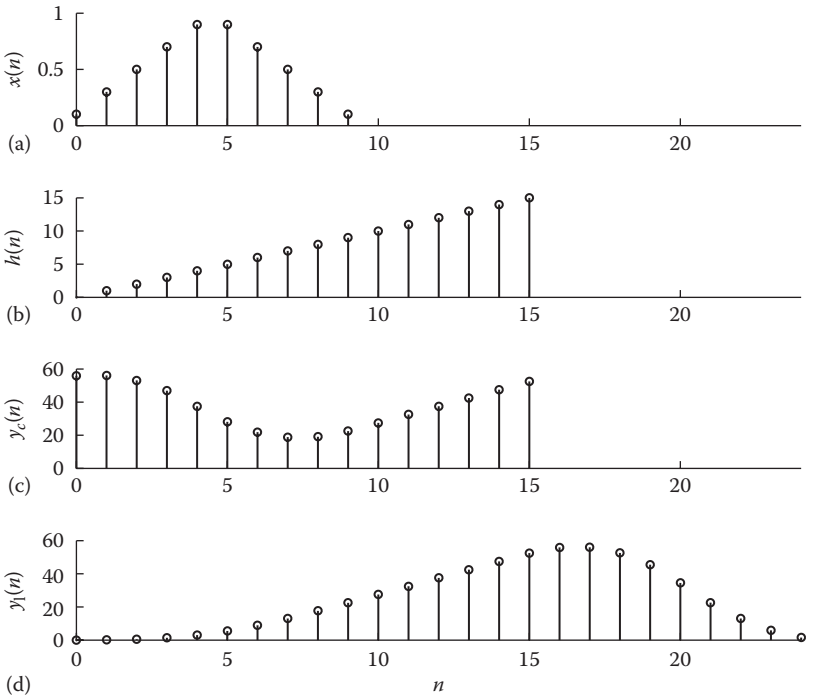


FIGURE 2.3 (a) A triangle sequence $x(n)$ and (b) a truncated ramp sequence $h(n)$. (c) The circular convolution of $x(n)$ and $h(n)$ and (d) the linear convolution of the two sequences.

2.4.3 Windowing

The definition of the DTFT (Equation 2.33) assumes knowledge of the values of $x(n)$ for all n. In practice, we only observe $x(n)$ for a finite duration, say, for N samples. Mathematically, this is equivalent to truncating $x(n)$ via the use of a finite-duration function, which we call a *window*. The signal truncation can be expressed mathematically as

$$x_W(n) = x(n)w(n) \tag{2.54}$$

where

$w(n)$ denotes the window function
$x_W(n)$ denotes the truncated signal

One simple window function to use is the *rectangular function* defined as

$$w(n) = \begin{cases} 1 & 0 \leq n \leq L-1 \\ 0 & \text{otherwise} \end{cases} \tag{2.55}$$

As we will see shortly, the windowing operation in Equation 2.54 affects the spectrum of $x(n)$ in many ways, depending on the shape of the window function. To demonstrate this, we consider in the following example the DTFT of a truncated sinusoid.

Example 2.2:

Suppose that the discrete-time signal $x(n)$ is a single sinusoid given by

$$x(n) = 2\cos(\circ\ {}_0 n) \tag{2.56}$$

where $\omega_0 = 2\pi 0.1$. Compute the DTFT of $x_W(n)$ given in Equation 2.54, where $w(n)$ denotes the window function given in Equation 2.55.

Using Euler's formula, we can express $x(n)$ as

$$x(n) = e^{j2\pi 0.1n} + e^{-j2\pi 0.1n} \tag{2.57}$$

After multiplying $w(n)$ in Equation 2.55 by the exponentials given in the preceding text and using the modulation property (Equation 2.44) of the DTFT, we obtain the DTFT of the windowed signal $x_W(n)$ as follows:

$$X_W(\omega) = W(\omega - \omega_0) + W(\omega + \omega_0) \tag{2.58}$$

where $W(\omega)$ is the DTFT of $w(n)$. Using the result from Example 2.1 along with the time-shifting property, we can compute $W(\omega)$ as follows:

$$W(\omega) = e^{-j\omega(L-1)/2}\left[\frac{\sin(\omega L/2)}{\sin(\omega/2)}\right] \tag{2.59}$$

Note that the first zero crossing of $W(\omega)$ occurs at $\omega = 2\pi/L$, which is regarded as the effective bandwidth of $W(\omega)$ (see Figure 2.4). Figure 2.5 shows the magnitude spectrum, $|X_W(\omega)|$, obtained by computing the N-point DFT of $x_W(n)$, where $N > L$, and L is the duration of the window. Figure 2.5 shows the magnitude spectrum using $N = 1024$ and different values of L, that is, $L = 100$, 300, and 500. As can be seen, $|X_W(\omega)|$ has a single peak at the frequency of the sinusoid, that is, at $f_0 = 0.1$. The spectrum, however, is not narrowband or localized near the signal frequency but, rather, it is spread out over the entire frequency range ($-0.5 < f < 0.5$). This phenomenon, caused by the windowing, is called *spectral leakage*, as information from the signal frequency leaks out to other frequencies. Note that the true magnitude spectrum of a discrete sinusoid defined over $-\infty < n < \infty$ consists of two discrete impulses at the signal frequency, that is,

$$X(\omega) = \delta(\omega - \omega_0) + \delta(\omega + \omega_0) \tag{2.60}$$

The true magnitude spectrum is thus completely localized at the signal frequency. From Figure 2.5, we can see that as L gets larger (i.e., window gets longer in duration), the spectrum gets more localized, resembling that of two discrete impulses (Equation 2.60). This is because the effective bandwidth of $W(\omega)$ (Equation 2.59) becomes smaller as L becomes larger.

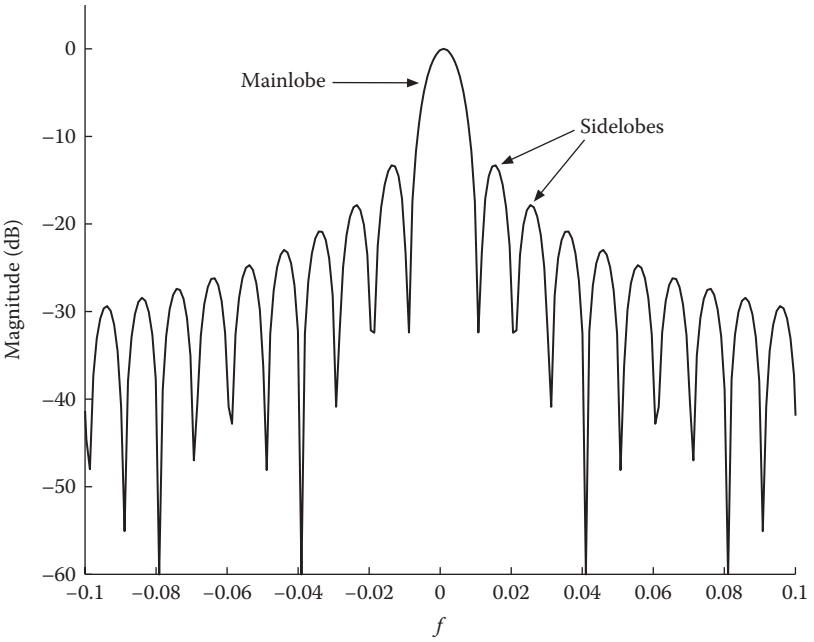


FIGURE 2.4 Magnitude spectrum of a rectangular window of length L.

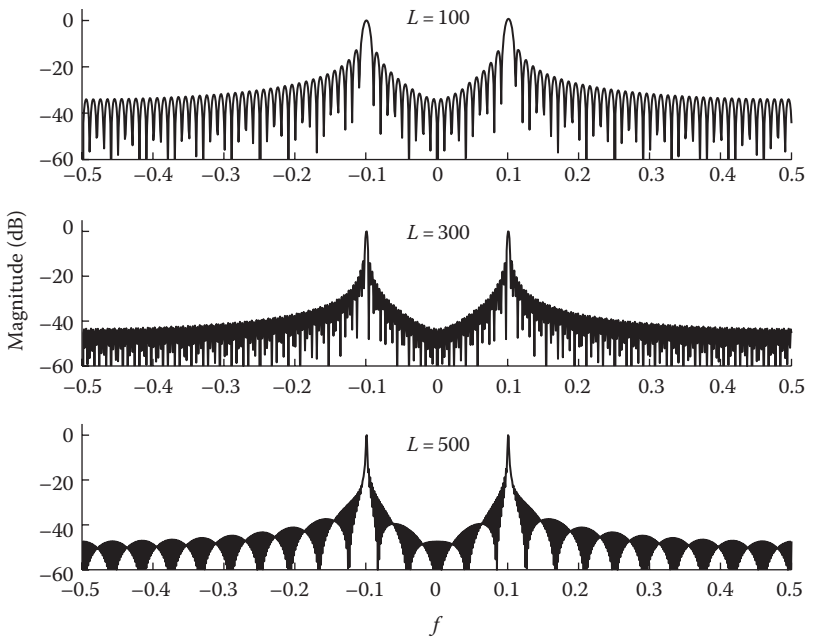


FIGURE 2.5 Spectra of a rectangular-windowed sine wave for different values of window length L.

The example given in the preceding text considered a rectangular window (Equation 2.55). A different picture emerges when a *Hamming window* is used. The Hamming window is defined as

$$w(n) = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{L-1}\right) & 0 \le n \le L-1 \\ 0 & \text{Otherwise} \end{cases} \tag{2.61}$$

Figure 2.6 shows the magnitude spectrum of the Hamming-windowed signal for different values of L. Compared to Figure 2.5, the spectral peak is broader (i.e., less localized); however, the spectral sidebands (also called *sidelobes*) are smaller. In fact, the bandwidth of the spectrum of the Hamming window is twice that of the rectangular window [3, p. 44]. This is more evident with $L = 500$. Hence, from Figures 2.5 to 2.6 we see that the spectrum of windowed signals is greatly affected by the shape of the windowing function.

Windowing not only alters the shape of the spectrum owing to the leakage effect, but also reduces the spectral resolution. The leakage effect reduces the ability to resolve or detect a weak (in amplitude) sine wave in the presence of a dominant (large-amplitude) sine wave. To illustrate this problem, we consider the following signal consisting of two sine waves

$$x(n) = 2\cos(\omega_1 n) + 0.1\cos(\omega_2 n) \tag{2.62}$$

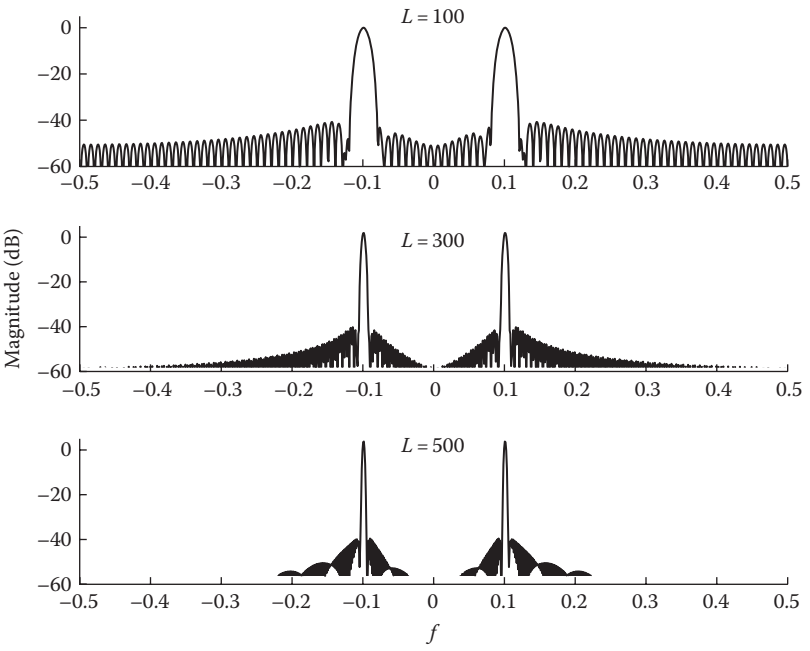


FIGURE 2.6 Spectra of a Hamming-windowed sine wave for different values of window length L.

where $\omega_1 = 0.1\pi$ and $\omega_2 = 0.11\pi$. The spectrum of the windowed sequence, truncated to L samples, is given by

$$X_W(\omega) = W(\omega - \omega_1) + W(\omega + \omega_1) + \frac{0.1}{2}[W(\omega - \omega_2) + W(\omega + \omega_2)] \qquad (2.63)$$

Figure 2.7 shows the magnitude spectrum $|X_W(\omega)|$ obtained using a rectangular window, and Figure 2.8 shows the spectrum obtained using a Hamming window for $L = 300$. In theory, there should be two distinct spectral lines centered at ω_1 and ω_2. However, as shown in Figure 2.7, the two spectral lines for the sequence truncated with a rectangular window are not distinguishable. In contrast, the two spectral lines are resolvable for the sequence truncated with the Hamming window. This is because the second spectral line is "buried" under the sidelobe of the rectangular window but remains above the first sidelobe of the Hamming window. Note that the first sidelobe of the spectrum of the rectangular window is −13 dB below the peak [3, p. 44]. However, the amplitude of the weak sine wave is −26 dB (=20 log (0.1/2)) below the peak and is therefore not detectable.

In brief, the windowing function needs to be chosen carefully, depending on the application. The selection of the window should be based on the trade-off between spectral smearing and leakage effects. For the previous example in which we considered the detection of a dominant sinusoid in the presence of a weak amplitude sinusoid, the Hamming windowing function proved to be a better choice. Aside from the rectangular and Hamming windows, other windowing functions can be used [3, Chapter 2].

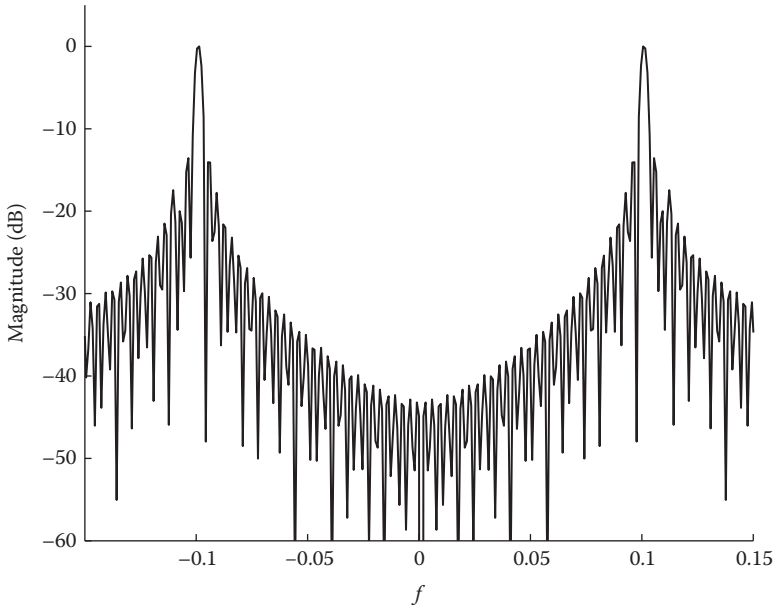


FIGURE 2.7 Spectrum of the sum of two sine waves $[x(n) = 2\cos(\omega_1 n) + 0.1\cos(\omega_2 n)]$ truncated using a rectangular window.

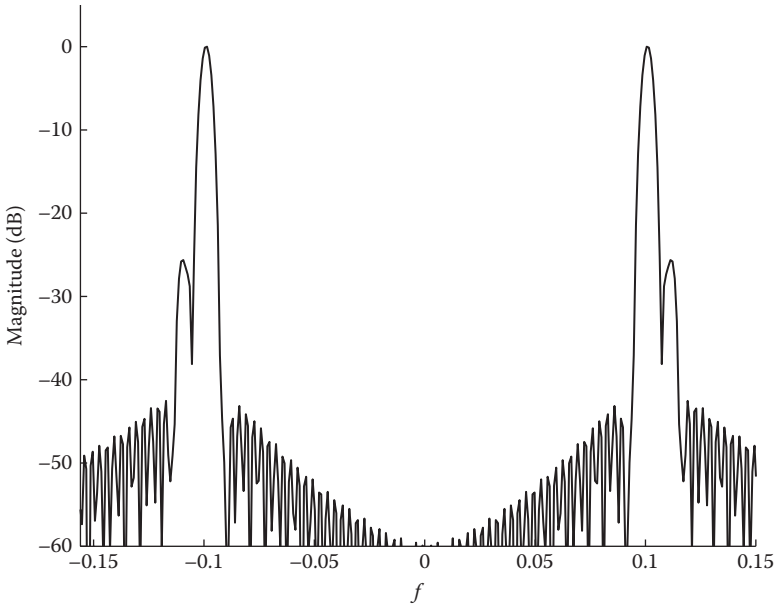


FIGURE 2.8 Spectrum of the sum of two sine waves [$x(n) = 2\cos(\omega_1 n) + 0.1\cos(\omega_2 n)$] truncated using a Hamming window.

2.5 SHORT-TIME FOURIER TRANSFORM

In the previous section, we presented the DTFT. This Fourier transform representation is more appropriate for stationary or deterministic signals and not necessarily for speech signals with temporal and spectral characteristics changing markedly over time. The DTFT representation is, however, applicable to speech processing, provided we consider analysis of short segments (10–30 ms) of speech, during which the properties of speech do not change much.

In this section, we introduce a time-varying Fourier representation that reflects the time-varying properties of the speech waveform. In-depth coverage of the short-time Fourier transform (STFT) can be found in [4, Ch. 7; 5, Ch. 6; 6].

2.5.1 Definition

The STFT is given by

$$X(n,\omega) = \sum_{m=-\infty}^{\infty} x(m)w(n-m)\, e^{-j\omega m} \tag{2.64}$$

where $x(m)$ is the input signal and $w(m)$ is the analysis window, which is time-reversed and shifted by n samples (see example in Figure 2.9). The STFT is a function of two variables: the discrete-time index, n, and the (continuous) frequency variable, ω. To obtain $X(n + 1, \omega)$, we slide the window by one sample, multiply it with $x(m)$, and

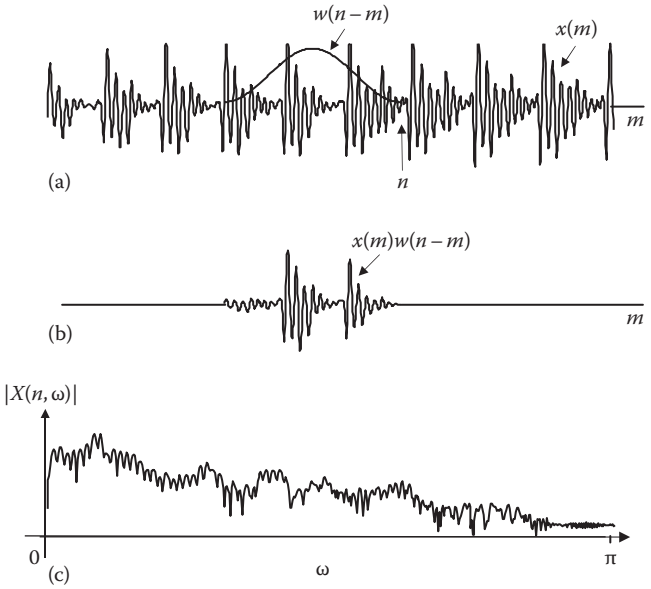


FIGURE 2.9 Steps involved in the computation of the short-time Fourier transform, $X(n, \omega)$: (a) the window (Hamming window, in this example) shifted by n samples, (b) the windowed signal, and (c) the magnitude spectrum of the windowed signal.

compute the Fourier transform of the windowed signal. Continuing this way, we generate a set of STFTs for various values of n until we reach the end of the signal $x(m)$. Note that the two-dimensional function $|X(n, \omega)|^2$ provides the so-called spectrogram of the speech signal—a two-dimensional display of the power spectrum of speech as a function of time. This is a widely used tool employed for studying the time-varying spectral and temporal characteristics of speech (more on this in Section 2.6).

Figure 2.9 shows the steps involved in the computation of the STFT, $X(n, \omega)$. Figure 2.9a shows the window (Hamming window, in this example) shifted by n samples, Figure 2.9b shows the windowed signal, and Figure 2.9c shows the magnitude spectrum of the windowed signal.

Similar to the discrete Fourier transform derivation, one can obtain a discrete version of the STFT by sampling the frequency variable ω at N uniformly spaced frequencies, that is, at $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$. The resulting discrete STFT is defined as

$$X(n, \omega_k) \triangleq X(n, k) = \sum_{m=-\infty}^{\infty} x(m) w(n-m)\, e^{-j\frac{2^\circ}{N}km} \tag{2.65}$$

The STFT $X(n, \omega)$ can be interpreted in two distinct ways, depending on how we treat the time (n) and frequency (ω) variables. If, for instance, we assume that n is fixed but ω varies, then $X(n, \omega)$ can be viewed as the Fourier transform of a windowed sequence. If we assume that ω is fixed and the time index n varies, a filtering interpretation emerges.

2.5.2 Interpretations of the STFT

We begin with the Fourier transform interpretation in which the time index n is held fixed. When comparing the equation of the DTFT (Equation 2.33) with the equation of the STFT (Equation 2.64) for a fixed value of n, we observe that $X(n,\omega)$ is the normal DTFT of the sequence $x(n-m)w(m)$. As such, $X(n,\omega)$ has the same properties as the DTFT.

The STFT can also be viewed as the output of a filtering operation. For this interpretation, we fix the value of ω at, say, ω_k and vary the time index n. We can rewrite Equation 2.64 as the convolution of two sequences:

$$X(n,\omega_k) = \sum_{m=-\infty}^{\infty} \left[x(m)e^{-j\omega_k m} \right] \cdot w(n-m)$$

$$= \left[x(n)e^{-j\omega_k n} \right] * w(n) \tag{2.66}$$

The STFT at frequency ω_k can be viewed as the output of a system with input $x(n)e^{-j^{\circ}{}_k n}$ and impulse response $w(n)$. The analysis window $w(n)$ plays here the role of the filter's impulse response and is sometimes called the *analysis filter*. The signal $x(n)$ is first multiplied by $e^{-j^{\circ}{}_k n}$ (corresponding to a shift in the frequency domain by ω_k), and the output signal is passed through a filter whose impulse response is $w(n)$ (see Figure 2.10). After making use of the modulation (Equation 2.44) and convolution (Equation 2.47) properties of the DTFT, we can express Equation 2.66 in the frequency domain in terms of the individual DTFTs of the two signals, that is,

$$X(n,\omega_k) = X(\omega+\omega_k)W(\omega) \tag{2.67}$$

where

$W(\omega)$ is the DTFT of $w(n)$

$X(\omega)$ is the DTFT of $x(n)$

This filtering operation is shown graphically in Figure 2.10.

Yet another interpretation of the STFT can be obtained as follows. Letting $m' = n - m$ in Equation 2.64 and changing the summation indices accordingly, we can express $X(n, \omega)$ as follows:

$$X(n,\omega) = \sum_{m=-\infty}^{\infty} x(n-m)w(m)\, e^{-j\omega(n-m)}$$

$$= e^{-j\omega n} \sum_{m=-\infty}^{\infty} x(n-m)w(m)\, e^{j\omega m} \tag{2.68}$$

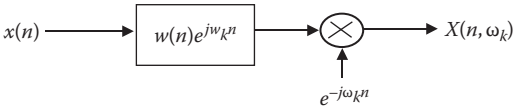


FIGURE 2.10 Variant filtering interpretation of the short-time Fourier transform.

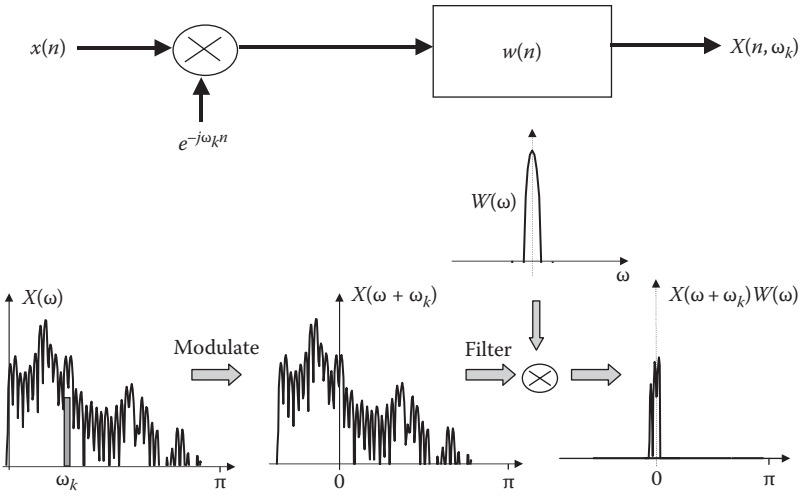


FIGURE 2.11 Filtering interpretation of the short-time Fourier transform.

Realizing that the summation term in the previous equation is nothing but the convolution of the signal $x(n)$ with $w(n)e^{-j^{\circ}{}_{k}n}$, we can express $X(n, \omega)$ at $\omega = \omega_k$ as

$$X(n, \omega_k) = e^{-j\omega_k n}\left[x(n) * w(n)e^{j\omega_k n}\right] \tag{2.69}$$

This filtering operation is depicted in Figure 2.11.

2.5.3 Sampling the STFT in Time and Frequency

The STFT $X(n, \omega)$ is a two-dimensional function of time n and frequency ω. In principle, $X(n, \omega)$ can be evaluated for each value of n; however, in practice, $X(n, \omega)$ is decimated in time due partly to the heavy computational load involved and partly to the redundancy of information contained in consecutive values of $X(n, \omega)$ (e.g., between $X(n, \omega)$ and $X(n + 1^{\circledR}\omega)$). Hence, in most practical applications $X(n, \omega)$ is not evaluated for every sample but for every R sample, where R corresponds to the decimation factor, often expressed as a fraction of the window length. The sampling, in both time and frequency, has to be done in such a way that $x(n)$ can be recovered from $X(n, \omega)$ without aliasing.

We first consider the sampling of $X(n, \omega)$ in time. For that, we will use the filtering interpretation of the STFT. It is clear from Equation 2.67 and Figure 2.10 that the bandwidth of the sequence $X(n, \omega_k)$ (along n, for a fixed frequency ω_k) is less than or equal to the bandwidth of the analysis window $w(n)$. This suggests that $X(n, \omega_k)$ has to be sampled at twice the bandwidth of the window $w(n)$ to satisfy the Nyquist sampling criterion. If, for instance, the effective bandwidth of the analysis window is B Hz, then according to the Nyquist theorem $X(n, \omega_k)$ has to be sampled at a rate of $2B$ samples/second.

Let us consider, for example, the L-point Hamming window, which has an effective bandwidth of [3, p. 44]

$$B = \frac{2F_s}{L}\text{Hz} \tag{2.70}$$

where F_s is the sampling frequency. For this window, $X(n, \omega_k)$ has to be sampled in time at a minimum rate of $2B$ samples/second = $4F_s/L$ samples/second to avoid time aliasing. The corresponding sampling period is $L/(4F_s)$ second or $L/4$ samples. This means that for an L-point Hamming window, $X(n, \omega_k)$ needs to be evaluated at most every $L/4$ samples, corresponding to a minimum overlap of 75% between adjacent windows. This rather strict requirement on the minimum amount of overlap between adjacent windows can be relaxed if we allow zeros in the window transform [4, Ch. 7]. In speech enhancement applications, it is quite common to use a 50% rather than 75% overlap between adjacent windows. This implies that $X(n, \omega_k)$ is evaluated every $L/2$ samples; that is, it is decimated by a factor of $L/2$, where L is the window length.

The sampling requirement in the frequency domain can be derived using the Fourier transform interpretation of the STFT. We know that the STFT $X(n, \omega_k)$ (for fixed n) is the DTFT of the windowed sequence $w(m)x(n - m)$. Hence, to recover the windowed sequence $w(m)x(n - m)$ with no aliasing, we require that the frequency variable ω be sampled at N ($N \geq L$) uniformly spaced frequencies, that is, at $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N - 1$.

2.5.4 Short-Time Synthesis of Speech

Given $X(n, \omega)$, it is necessary to reconstruct the time sequence $x(n)$. This is represented mathematically by a *synthesis equation* that expresses $x(n)$ in terms of its STFT. Consider, for example, the Fourier transform interpretation of the STFT. The fact that $X(n, \omega)$ is the DTFT of the windowed sequence $w(n - m)x(m)$ (for a fixed value of n) implies that the sequence $w(n - m)x(m)$ can be recovered exactly via the use of the inverse DTFT, that is,

$$w(n-m)x(m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(n,\omega)e^{j\omega n}\, d\omega \tag{2.71}$$

Assuming $w(0) \neq 0$ and after evaluating Equation 2.71 at $m = n$, we obtain

$$x(m) = \frac{1}{2\pi w(0)}\int_{-\pi}^{\pi} X(m,\omega)e^{j\omega m}\, d\omega \tag{2.72}$$

This equation clearly demonstrates that it is possible to recover $x(n)$ exactly from its STFT. Equation 2.72 represents one of many synthesis equations for the STFT. This equation is inefficient, however, in that it requires that the analysis window slides one

sample at a time. In practice, more efficient methods can be used, which impose certain constraints on the properties of the windows as well as the time and frequency sampling resolution of $X(n, \omega)$.

We describe next two classical methods, the filterbank summation method and the overlap-add method [7–9], for short-time synthesis of speech. Of the two methods, the overlap-add method is typically used in speech enhancement. Other methods based on least-squares estimates of $x(n)$ from its STFT were proposed in [10].

2.5.4.1 Filterbank Summation for Short-Time Synthesis of Speech

This synthesis method is based on the filterbank interpretation of the STFT. From Equation 2.68, we have

$$X(n, \omega_k) = e^{-j\omega_k n} \sum_{m=-\infty}^{\infty} x(n-m) w_k(m) e^{j\omega_k m} \tag{2.73}$$

where $w_k(m)$ denotes the window applied at frequency ω_k. Defining $h_k(n)$ as

$$h_k(n) = w_k(n) e^{j^{\circ}{}_k n} \tag{2.74}$$

we can write Equation 2.73 as

$$\begin{aligned} X(n, \omega_k) &= e^{-j\omega_k n} \sum_{m=-\infty}^{\infty} x(n-m) h_k(m) \\ &= e^{-j\omega_k n} y_k(n) \end{aligned} \tag{2.75}$$

where $y_k(n)$ is defined as follows:

$$y_k(n) = \sum_{m=-\infty}^{\infty} x(n-m) h_k(m) \tag{2.76}$$

We observe from Equation 2.76 that $y_k(n)$ can be viewed as the output of a system with impulse response $h_k(n)$ at frequency ω_k (see Figure 2.12). From Equation 2.75, $y_k(n)$ can also be expressed as

$$y_k(n) = X(n, {}^{\circ}{}_k) e^{j^{\circ}{}_k n} \tag{2.77}$$

and the reconstructed signal is formed as

$$\begin{aligned} y(n) &= \sum_{k=0}^{N-1} y_k(n) \\ &= \sum_{k=0}^{N-1} X(n, \omega_k) e^{j\omega_k n} \end{aligned} \tag{2.78}$$

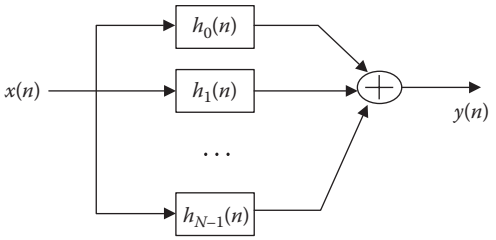


FIGURE 2.12 Filtering interpretation of the filterbank summation method for short-time speech synthesis.

where N is the number of uniformly spaced frequencies. The filtering interpretation of Equation 2.78 is shown in Figure 2.12. Note that Equation 2.78 gives the equation of the inverse discrete Fourier transform of $X(n, \omega_k)$ provided that $X(n, \omega_k)$ is properly sampled in frequency at N uniformly spaced frequencies, that is, at $\omega_k = 2\pi k/N$, $k = 0, 1, \ldots, N-1$, where $N \geq L$, and L is the window duration.

To find the relationship between $x(n)$ and $y(n)$, we consider the filtering operation in Figure 2.12 in the frequency domain. From Figure 2.12 and after using the distributive property of the convolution (Equation 2.19), we have

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) \\ &= X(\omega) \cdot \sum_{k=0}^{N-1} H_k(\omega_k) \end{aligned} \tag{2.79}$$

where $H_k(\omega)_k$ is the frequency response of the filter $h_k(n)$.

From Equation 2.74 we see that if $w_k(n)$ is the impulse response of a low-pass filter with cutoff frequency ω_c, then the frequency response of the filter $h_k(n)$ will be (due to the modulation property) that of a band-pass filter with center frequency ω_k and bandwidth $2\omega_c$. If we consider N such band-pass filters, then we will be able to cover the entire signal bandwidth. The frequency response of the kth band-pass filter, $H_k(\omega)$, is then given by

$$H_k(\omega) = W_k(\omega - \omega_k) \tag{2.80}$$

where $W_k(\omega)$ is the Fourier transform of $w_k(n)$. Hence, $H_k(\omega)$ is equal to the frequency response of the low-pass filter $W_k(\omega)$ shifted at frequency ω_k (the shift comes from the modulation property of the Fourier transform). If we assume that the same window $w_k(n)$ is used for all frequencies, that is, $w_k(n) = w(n)$ for all k, and that $W(\omega)$ is properly sampled in frequency (i.e., $N \geq L$, where L is the time duration of the window), then it can be shown that [5, p. 269]

$$\frac{1}{N} \sum_{k=0}^{N-1} W(\omega - \omega_k) = w(0) \tag{2.81}$$

Substituting Equations 2.80 and 2.81 into Equation 2.79, we finally get

$$Y(\omega) = X(\omega)N \cdot w(0) \tag{2.82}$$

or equivalently, $y(n) = x(n)N \cdot w(0)$. Hence, provided that the window $w(n)$ has a finite duration L (and $N \geq L$), we can reconstruct the signal $x(n)$ exactly from its STFT $X(n, \omega)$. This can be done independently of the shape of the window $w(n)$.

2.5.4.2 Overlap-and-Add Method for Short-Time Synthesis

An alternative method for reconstructing $x(n)$ from its STFT is the overlap-and-add method, which is widely used in speech enhancement.

Assume that the STFT $X(n, \omega)$ is sampled in time every R samples. We denote this sampling in the time dimension by $X(rR, \omega)$. The overlap-and-add method is motivated by the discretized version of Equation 2.72 and is based on the following equation [4, p. 325; 5, p. 275]:

$$y(n) = \sum_{r=-\infty}^{\infty} \left[\frac{1}{N} \sum_{k=0}^{N-1} X(rR, \omega_k) e^{j\omega_k n} \right] \tag{2.83}$$

The term in brackets (see also Equation 2.78) is an inverse discrete Fourier transform yielding for each value of r the sequence

$$y_r(n) = x(n)w(rR - n) \tag{2.84}$$

and therefore, Equation 2.83 can be expressed as

$$y(n) = \sum_{r=-\infty}^{\infty} y_r(n) = x(n) \sum_{r=-\infty}^{\infty} w(rR - n) \tag{2.85}$$

From Equation 2.85 we see that the signal $y(n)$ at time n is obtained by summing all the sequences $y_r(n)$ that overlap at time n. Provided that the summation term in Equation 2.85 is constant for all n, we can recover $x(n)$ exactly (within a constant) as

$$y(n) = C \cdot x(n) \tag{2.86}$$

where C is a constant. It can be shown that if $X(n, \omega)$ is sampled properly in time, that is, R is small enough to avoid time aliasing (see Section 2.5), then C is equal to [5, p. 275]

$$C = \sum_{r=-\infty}^{\infty} w(rR - n) = \frac{W(0)}{R} \tag{2.87}$$

independent of the time n. Equations 2.86 and 2.87 indicate that $x(n)$ can be reconstructed exactly (within a constant) by adding overlapping sections of the windowed sequences $y_r(n)$. The constraint imposed on the window is that it satisfies Equation 2.87; that is, the sum of all analysis windows shifted by increments of R samples adds up to a constant. Furthermore, R needs to be small enough to avoid time aliasing. For an L-point Hamming window, R has to be set to, $R = L/4$ suggesting that adjacent windows overlap by 75%. According to Equation 2.85, if $R = L/4$ the signal $y(n)$ consists of four terms:

$$y(n) = x(n)w(R-n) + x(n)w(2R-n) + x(n)w(3R-n) + x(n)w(4R-n) \quad (2.88)$$

where $0 \le n \le R-1$. With $R = L/2$ (i.e., 50% window overlap), which is most commonly used in speech enhancement, the signal $y(n)$ consists of two terms:

$$y(n) = x(n)w(R-n) + x(n)w(2R-n) \quad (2.89)$$

for $0 \le n \le R-1$. Figure 2.13 shows how the overlap addition is implemented for an L-point Hamming window with 50% overlap ($R = L/2$). In the context of speech enhancement, the enhanced output signal in frame t consists of the sum of the windowed signal [with $w(R-n)$] enhanced in the previous frame $(t-1)$ and the windowed signal [with $w(2R-n)$] enhanced in the present frame (t).

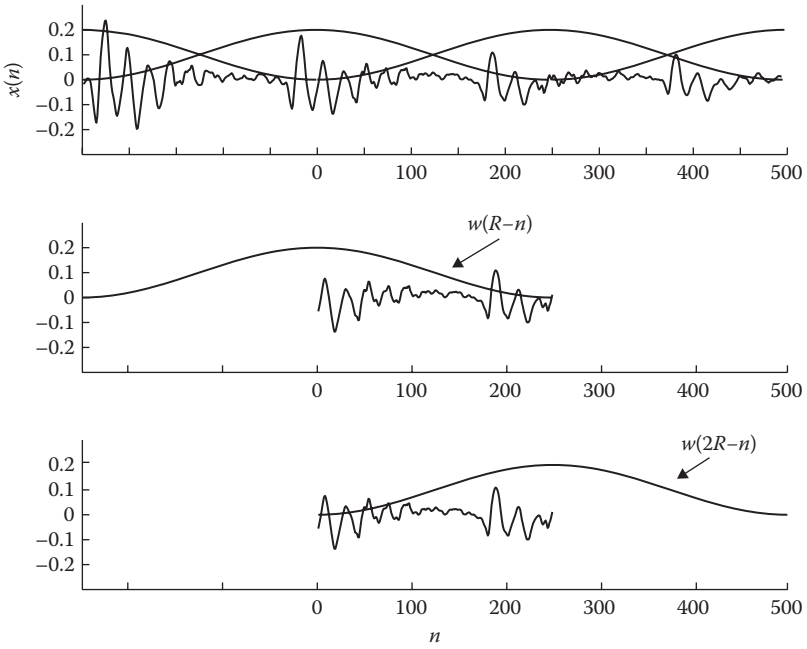


FIGURE 2.13 Example of reconstruction procedure for overlap-and-add method with 50% overlap. Window length is 500 samples ($L = 500$) and $R = L/2$.

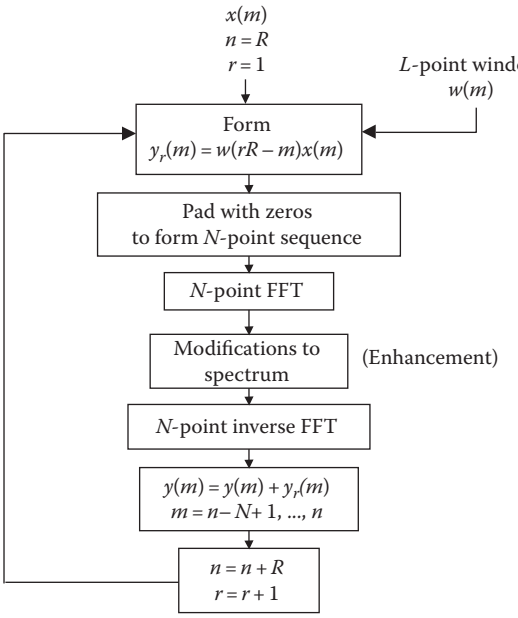


FIGURE 2.14 Flow diagram of overlap-and-add method.

Figure 2.14 shows the flow diagram of the overlap-and-add method, which can be used in any frequency domain speech enhancement algorithm. Note that the L-point signal sequence needs to be padded with sufficient zeros to avoid time aliasing. In the context of speech enhancement, the input signal $x(n)$ in Figure 2.13 corresponds to the noisy signal and the output signal $y(n)$ to the enhanced signal.

2.6 SPECTROGRAPHIC ANALYSIS OF SPEECH SIGNALS

The STFT is often used for spectrographic analysis of the speech signals. The *spectrogram* is a graphical display of the power spectrum of speech as a function of time and is given by

$$S(n,\omega) = \left|X(n,\omega)\right|^2 \tag{2.90}$$

where $X(n,\omega)$ denotes the STFT of the speech signal $x(n)$. The quantity $S(n,\omega)$ can be viewed as a two-dimensional "power-spectral density," the second dimension being time. The spectrogram describes the speech signal's relative energy concentration in frequency as a function of time and, as such, it reflects the time-varying properties of the speech waveform. Spectrograms are typically displayed in gray scale (see example in Figure 2.15). The larger the energy in the spectrum is at a specific frequency, the darker the display is at that frequency. Large magnitudes in the spectrum such as peaks are displayed in black or dark colors, whereas valleys are displayed in white colors (see example in Figure 2.15). Values falling between peaks and valleys are displayed using different shades of gray.

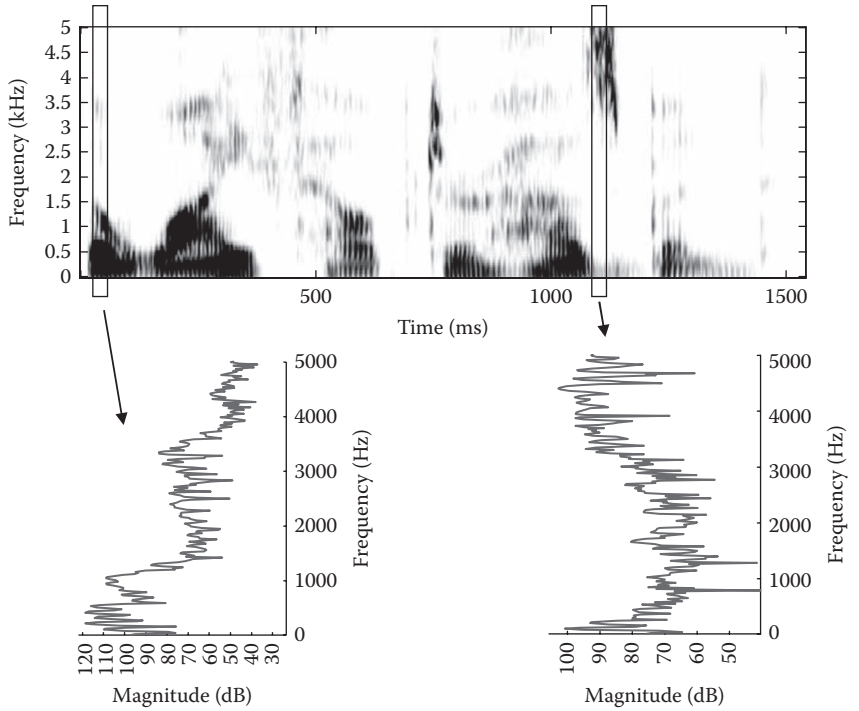


FIGURE 2.15 Spectrogram of the sentence "The wife helped her husband" produced by a male speaker. Bottom panels show the magnitude FFT spectrum obtained at two instances in time enclosed by the rectangles in the spectrogram.

Two kinds of spectrograms, *narrowband* and *wideband,* can be produced, depending on the window length used in the computation of $S(n, \omega)$. A long-duration window (at least two pitch periods long) is typically used in the computation of the narrowband spectrogram and a short window in the computation of the wideband spectrogram. The narrowband spectrogram gives good frequency resolution but poor time resolution. The fine frequency resolution allows the individual harmonics of speech to be resolved. These harmonics appear as horizontal striations in the spectrogram (see example in Figure 2.16a). The main drawback of using long windows is the possibility of temporally smearing short-duration segments of speech, such as the stop consonants. The wideband spectrogram uses short-duration windows (less than a pitch period) and gives good temporal resolution but poor frequency resolution. The main consequence of the poor frequency resolution is the smearing (in frequency) of individual harmonics in the speech spectrum, yielding only the spectral envelope of the spectrum (see example in Figure 2.16b). The wideband spectrogram provides good temporal resolution, making it suitable for analysis and examination of all sounds in the English language. The wideband spectrogram is an invaluable tool for speech analysis, and we will be using it throughout the text.

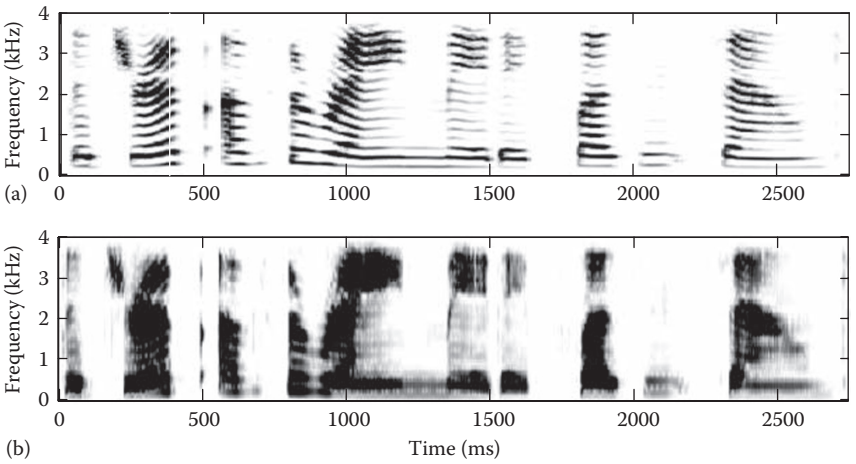


FIGURE 2.16 (a) Narrowband and (b) wideband spectrograms of the sentence "The drip of the rain made a pleasant sound" produced by a female speaker.

2.7 SUMMARY

This chapter provided a review of the fundamentals of discrete-time signal processing (a more thorough coverage of signal processing principles can be found in [1,2]). We reviewed discrete-time signals and linear time-invariant systems, along with their z-transform and Fourier transform representations. The concept of windowing was covered, as most enhancement algorithms operate on a frame-by-frame basis rather than on a sample-by-sample basis. We introduced the STFT as a more suitable tool for representing the speech signal with temporal and spectral characteristics changing markedly over time. We mentioned that the DTFT representation is still applicable to speech processing provided we consider analysis of short segments (10–30 ms) of speech, during which the properties of speech do not change much. We presented the overlap-and-add method as a common synthesis approach used by nearly all speech enhancement algorithms. Lastly, we introduced the spectrogram (derived from the STFT) as a valuable tool that can be used to obtain a time-frequency representation of speech.

REFERENCES

1. Proakis, J. and Manolakis, D. (1996), *Digital Signal Processing*, 3rd ed., Upper Saddle River, NJ: Prentice Hall.
2. Oppenheim, A., Schafer, R., and Buck, J. (1999), *Discrete-Time Signal Processing*, 2nd ed., Upper Saddle River, NJ: Prentice Hall.
3. Stoica, P. and Moses, R. (1997), *Introduction to Spectral Analysis*, Upper Saddle River, NJ: Prentice Hall.
4. Quatieri, T. (2002), *Discrete-Time Speech Signal Processing*, Upper Saddle River, NJ: Prentice Hall.
5. Rabiner, L. and Schafer, R. (1978), *Digital Processing of Speech Signals*, Englewood Cliffs, NJ: Prentice Hall.

6. Nawab, S. and Quatieri, T. (1987), Short-time Fourier transform, in Lim, J. and Oppenheim, A. (Eds.), *Advanced Topics in Signal Processing*, Englewood Cliffs, NJ: Prentice Hall.
7. Allen, J. and Rabiner, L. (1977), A unified theory of short-time spectrum analysis and synthesis, *Proc. IEEE*, 65(11), 1558–1564.
8. Portnoff, M. (1980), Time-frequency representation of digital signals and systems based on short-time Fourier analysis, *IEEE Trans. Acoust. Speech Signal Process*., 28(1), 55–69.
9. Crochiere, R. E. (1980), A weighted overlap-add method of short-time Fourier analysis/synthesis, *IEEE Trans. Acoust. Speech Signal Process*., 28(1), 99–102.
10. Griffin, D. and Lim, J. (1984), Signal estimation from modified short-time Fourier analysis/synthesis, *IEEE Trans. Acoust. Speech Signal Process*., 32(2), 236–243.